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21 November 1980

MILITARY SPECIFICATION

CONTROL, RADIO SET C-6709( )/G

This specification is approved for use by the Communications Research and Development Command, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification covers one type of control equipment for use between automatic or semi-automatic analog switchboards and Single Channel Radio Sets, designated as Control, Radio Set C-6709( )/G. (See 6.1)

2. APPLICABLE DOCUMENTS

2.1 Issues of Documents. The following documents of the issue in effect on date of invitation for bid, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-P-116	Preservation-Packaging, Methods of
MIL-P-11268	Parts, Materials, and Processes used in Electronics Equipment
MIL-M-13231	Marking of Electronic Items
MIL-F-14072	Finishes for Ground Electronic Equipment
MIL-H-49092	Headset/Microphone, H-325( )/TTC

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: Cdr, US Army Communications Research and Development Command, ATTN: DRDCO-CM-DM, Fort Monmouth, NJ 07703 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 5820

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STANDARDS

MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects
MIL-STD-454	Standard General Requirements for Electronic Equipment
MIL-STD-461	Electromagnetic Interference Characteristics Requirements for Equipment Subsystems and Systems
MIL-STD-462	Electromagnetic Interference Characteristics, Measurement of
MIL-STD-810	Environmental Test Methods
MIL-STD-781	Reliability Tests Exponential Distribution

DRAWINGS/SPECIFICATIONS

DL-SM-B-938013	Control, Radio Set C-6709( )/G
DL-SM-B-317264	Headset-Microphone H-325/TTC
SM-D-415553	Cable Assembly CX-4723( )VRC

OTHER DOCUMENTS

Special Packaging Instruction #1G0062

3. REQUIREMENTS

3.1 First Article. When specified in the contract or purchase order, the contractor shall furnish sample units for First Article inspection and approval. (See 4.3).

3.2 Construction. The equipment shall be constructed in accordance with the requirements of this specification and Drawing and Data List DL-SM-B-938013, (including the requirements for parts, materials, and processes thereon). (See 4.4).

3.3 Parts, Materials and Processes, General. In addition to the requirements of this specification, the requirements of MIL-P-11268, including the selection requirements therein, shall apply. (See 4.4).

3.4 Finish. The equipment shall be finished in accordance with MIL-F-14072 and the equipement drawings. (See 4.4).

3.5 General. Marking shall conform to Specification MIL-M-13231, (See 4.4). Panel markings shall be Group I.

3.5.2 Visibility. Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of equipment. (See 4.4).

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3.5.3 Serial Numbers. The equipment shall be serially numbered.

3.6 Interchangeability. Dimensions of C-6709( )/G shall be as specified in the applicable drawings; and like units, assemblies, subassemblies, and replaceable parts shall meet Requirement 7 of MIL-STD-454. (See 4.14).

3.7 Preconditioning.

3.7.1 Bounce Preconditioning. Each equipment shall meet the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.6.1.

3.7.2 Electrical Preconditioning (Burn-In). Prior to submission to the specified quality conformance inspection, each unit shall be subjected to a minimum 48 hour on-time burn-in period under the conditions specified for reliability testing. The last 24 hours of the burn-in period shall be failure free. (See 4.6.2).

3.8 Electrical

3.8.1 Generated Signals.

3.8.1.1 Sieze. The C-6709( )/G shall generate a sieze signal of 2250 ± 29.3 Hz at a level of -7 dBm ± 2 dBm when the "SWB" button is activated. (See 4.8.1.1).

3.8.1.2 Release. The C-6709( )/G shall generate a release signal of 2600 ± 33.8 Hz at a level of -7 dBm ± 2 dB when the release button is activated. Release shall time out between 3 and 10 seconds if release acknowledge is not received by C-6709( )/G. (See 4.8.1.2).

3.8.1.3 Sieze/Release Acknowledge. The C-6709( )/G shall generate a signal of 570 ± 7.4 Hz at a level of -7 dBm ± 2 dB upon detection of sieze or release signal. Upon cessation of sieze or release signal, sieze/release acknowledge shall be cut off after 250 milliseconds. (See 4.8.1.3).

3.8.1.4 Ringback. The C-6709( )/G shall generate a signal of 425 ± 5 Hz at a level of -7 dBm ± dB when SVC/BUSY Lamp is flashing. (See 4.8.1.4).

3.8.1.5 DTMF Signaling.

3.8.1.5.1 Dual Keys. Depressing two keys simultaneously in each row or in each column of key pad on C-6709( )/G, the output level shall be -7 dBm ± 2 dB. Frequencies shall be as follows:

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Column	1	1209 Hz ± 15.7 Hz
	2	1336 Hz ± 17.4 Hz
Column	3	1477 Hz ± 19.2 Hz
	4	1633 Hz ± 21.2 Hz
Row	1	697 Hz ± 9.06 Hz
	2	770 Hz ± 10.1 Hz
	3	852 Hz ± 11.1 Hz
	4	941 Hz ± 12.2 Hz

3.8.2 Detection

3.8.2.1 Seize. The C-6709( )/G shall detect 2250 ± 45 Hz with signal power input from -5 dBm to -30 d Bm. Detection time shall be 700 ± 100 milliseconds. Detection shall not respond to signals beyond 5% of nominal frequency. (See 4.8.2.1).

3.8.2.2 Release. The C-6709( )/G shall detect 2600 ± 52 Hz with signal power input from -5 dBm to -30 dBm when in the idle or busy condition. Detection time in the idle condition shall be 325 ± 100 milliseconds. Detection time in the busy condition shall be 700 ± 100 milliseconds. Detectors shall not respond to signals beyond 6% of nominal frequency. (See 4.8.2.2).

3.8.2.3 Sieze/Release Acknowledge. The c-6709( )/G shall detect 570 ± 11.4 Hz with signal power input from -5 dBm to -30 dBm. Detection time shall be 450 ± 50 milliseconds. Detection shall not respond to signals beyond 5% of nominal frequency. (See 4.8.2.3).

3.8.2.4 Control Signal (DTMF)

3.8.2.4.1 Transmit On. The C-6709( )/G shall detect 697 ± 13.9 Hz/1209 ± 24.2 Hz with signal power input from -5 dBm to -30 dBm. Detection time shall be 150 ± 25 milliseconds. Detection shall not respond to signals beyond 5% of nominal frequency. (See 4.8.2.4.2).

3.8.2.4.2 Transmit Off. The C-6709( )/G shall detect 697 ± 13.9 Hz/1477 ± 29.6 Hz with signal power input from -5 dBm to -30 dBm. Detection time shall be 150 ± 25 milliseconds. Detection shall not respond to signals beyond 5% of nominal frequency. (See 4.8.2.4.2).

3.8.2.5 Control Signal (Single Frequency).

3.8.2.5.1 Transmit On. The C-6709( )/G shall detect 1209 ± 24.2 with signal power input from -5 dBm to -30 dBm. Detection time shall be 375 ± 50 milliseconds. Detection shall not respond to signals beyond 5% of nominal frequency. (See 4.8.2.5).

3.8.2.5.2 Transmit Off. The C-6709( )/G shall detect 1477 ± 29.6 Hz with signal power input from -5 dBm to -30 dBm. Detection time shall be 375 ± 50 milliseconds. Dectection shall not respond to signals beyond 5% of nominal frequency. (See 4.8.2.5).

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3.8.2.6 Control Signal (AVOX). Analog Voice Operated Control (AVOX) will be used to control the receiver/transmitter state. The AVOX Analog Energy Threshold Detector shall automatically adjust the detector sensitivity to a level 3 dB ± 1 dB above an ambient background noise. The detector shall turn the transmitter on and off when the voice signal exceeds the background noise level by 3 dB ± 1 dB. The time to activate the control line shall be within 200 milliseconds. The control line shall be deactivated when the signal energy is below the threshold for at least 500 milliseconds. The AVOX Analog Threshold Detector shall detect signals as low as -35 dBm ± 3 dB. (See 4.8.2.6).

3.8.3 Impedance

3.8.3.1 Wire Side. The C-6709( )/G shall present an impedance of 600 ohms ± 10% at an angle of ± 30° to the wire side (both transmit and receive). (See 4.8.3.1).

3.8.3.2 Radio Side. The C-6709( )/G shall present an impedance of 600 ohms ± 10% at an angle of ± 30° to the receive terminal with the radio level switch in position 1 through 3. With radio level switch in position 4 through 8 at the receive terminal, the presented impedance shall be 150 ohms ± 10% at an angle of ± 30°. Looking into the send terminal on radio side of the C-6709( )/G a current source of 5K ohms or greater impedance shall be presented when the radio level switch is in position 1 through 8. (See 4.8.3.2).

3.8.4 Frequency Response. On any connection (all radio level switch positions) through the C-6709( )/G the frequency attenuation characteristics at constant power level shall be flat within + 1 dB with respect to the 1000 Hz value over the range of 0.5 KHz to 70 KHz. From 0.3 KHz to 0.5 KHz the response shall be between ± 1 dB and -30 dB with respect to the 1000 Hz level. (See 4.8.4).

3.8.5 Longitudinal Balance. The Longitudinal Balance at the wire side of the circuit shall be 40 dB or greater over the frequency range 300 Hz to 50 KHz, and shall be 25 dB or greater from 50.1 KHz to 70 KHz. (See 4.8.5).

3.8.6 Amplitude Response. The amplitude response of the radio control set between the wire receive terminals and the transmitter shall conform to Figure 1, 2, and 3. The output to the transmitter shall not vary from these curves by more than ± 10% when terminated into 150 ohms ± 1% resistive load with the specified inputs to the wire receiver terminals. (See 4.8.6).

3.8.6.1 Audio Signals. The audio signals from the radio receiver shall be taken from the monitor output of the receiver-transmitter. The response of the radio control set shall be such that a signal of 0.3 ± 0.1 volts will be presented to the wire send terminals when a signal of .22 volts is received from the radio receiver in any radio level position. (See 4.8.6.1).

3.8.7 Operator's Handset/Headset. The C-6709( )/G shall be capable of operating with headset/microphone H-325/TTC or handset H-250( )/U.

3.8.7.1 Acoustic Output (Headset). The headset shall produce a minimum output of -10 dBm at the 4-wire send terminals with a 28 dynes/cm^2 1000 Hz acoustic input to the headset microphone. (See 4.8.7).

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3.8.7.2 Sidetone (Headset). The sidetone coupling loss between transmitter input and receiver output shall be 20 ± 4 dB. (See 4.8.7).

3.8.7.3 Handset Functions (PTT). The push-to-talk (PTT) function utilizing the H-250/U handset shall meet the following requirements (see 4.8.7):

a. With operator connected to the switchboard, PTT will allow operator to talk but not key the transmitter with mode switch in any position.

b. With operator connected to radio or cut-through and mode switch in "oper", PTT will key transmitter and allow operator to talk.

c. With operator in (cut-thru) and mode switch in any position other than "oper", PTT will not key the transmitter but will allow operator to talk.

d. Operator out, PTT is not functional.

3.8.8 Noise Level. The idle noise level of the C-6709( )/G shall not exceed 27 dBa, with F1A weighting at both output ports. (See 4.8.8).

3.8.9 Power Requirements. The C-6709( )/G shall operate with (a) a steadystate voltage in the range of 22 to 32 Vdc, (b) other rectifier power supplies having a nominal 24 volt dc output with a max 1% ac ripple. Power requirements shall not exceed 5 watts at 24 volts. (See 4.8.9).

3.8.9.1 Polarity Protection. The C-6709( )/G shall not sustain any damage, to include blowing fuses, when the input power polarity is reversed. (See 4.8.9.1).

3.8.10 Distortion. The harmonic distortion shall not exceed 3% in either direction. (See 4.8.10).

3.8.11 Audible and Visual Alarms.

3.8.11.1 Audible Alarms. The audible alarm shall produce a sound level of 70 dB above .0002 dynes/cm^2 at 3 feet and shall be linearly adjustable from full-on to full-off. (See 4.8.11.1).

3.8.11.2 Visual Alarm. The indicator shall flash upon detection of sieze tone and shall illuminate continuously when "call/answer" button is depressed. The indicator shall stay illuminated until the C-6709( )/G has concluded its release sequence with the switchboard. (See 4.8.11.2).

3.8.12 Operational Performance Test. The C-6709( )/G shall provide satisfactory performance when tested in accordance with para 4.8.12.

3.9 Environmental. The C-6709( )/G shall meet the following environmental conditions.

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3.9.1 Temperature. The C-6709( )/G shall show no physical damage following testing as specified in 4.7.1 and shall meet the requirements of 3.8.

3.9.1.1 Low Temperature. During and after testing as specified in 4.7.1.1, the equipment shall meet the requirements of 3.8.

3.9.1.2 High Temperature. During and after testing as specified in 4.7.1.2, the equipment shall meet the requirements of 3.8.

3.9.2 Humidity. During and after testing as specified in 4.7.3, the equipment shall exhibit no physical damage, such as corrosion, rust, blistering, swelling or deterioration of parts and materials, and shall meet the requirements of 3.8.

3.9.3 Altitude. The equipment shall show no evidence of damage or deterioration in performance at altitudes up to 15,000 feet above sea level and after storage at altitudes up to 40,000 feet above sea level. (See 4.7.2).

3.9.4 Immersion. The equipment shall show no evidence of leakage after testing as specified in 4.7.7.

3.9.5 Vibration. The equipment shall show no physical damage following testing as specified in 4.7.4.1, and shall meet the requirements of 3.8.

3.9.5.1 Bounce, Loose Cargo. Following testing as specified in 4.7.4.2, the equipment shall meet the requirements of 3.8 and shall sustain no physical damage which would impair operation.

3.9.5.2 Shock, Bench Handling. Following testing as specified in 4.7.5, the equipment shall meet the requirements of 3.8 and shall sustain no physical damage.

3.9.5.3 Shock, Drop Test. Following testing as specified in 4.7.6, the equipment shall meet the requirements of 3.8 and any physical damage shall be minor as specified in MIL-STD-252.

3.9.6 Dust. When tested in accordance with 4.7.8, the equipment shall meet the requirements of 3.8.

3.9.7 Salt Fog. After the salt fog test as specified in 4.7.9, the equipment when examined visually with the aid of 10 power magnifier, shall show no evidence of degradation, such as flaking, pitting, blistering or loosening of finish on metal surface, or exfoliation of metal. (See 6.8).

3.9.8 Rain. The equipment shall show no physical damage following testing as specified in 4.7.11 and shall meet the requirements of 3.8.

3.9.9 Fungus. After the fungus test of 4.7.10, the equipment shall be visually examined using a 10-power magnifier. The equipment shall show no more than six minute unrelated spots, each no greater than .015 square inches in area, of sparse microbial growth as evidenced by growth colonization which included branching and sporulation on or within each cubic foot, or fraction thereof, of equipment assembly volume. Isolated instances of partial tubular germination shall not be included in this evaluation. (See 6.8).

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3.10 Electromagnetic Interference (See 4.9).

3.10.1 Electromagnetic Compatibility. The C-6709( )/G shall comply with the emission and susceptibility requirements of MIL-STD-461A, Notice 4 as follows: (See 4.9.1)

CE0 1	CS01	RE02	RS03
CE03	CS02	RE02.1	RS03.1
CE04	CS06		
CE05			

The upper frequency unit of test for RE02.1 shall be 1 GHz.

3.10.2 Production Testing. For subsequent Group C production testing, only the following tests are required. (See 4.9.1).

CE04	50 KHz to 50 MHz
CE05	50 KHz to 50 MHz
RE02	2.0 MHz to 400 MHz
RE02.1	2.0 MHz to 400 MHz
RS03	2.0 MHz to 400 MHz

3.10.3 Bonding and Grounding. The DC resistance for all bonds and grounds (including connector shell to the case or chassis) shall not exceed 2.5 milliohms. (See 4.10).

3.11 Air Seal Test. The C-6709( )/G shall be subjected to the air seal test, and shall not exceed 0.01 pound/minute decrease in vacuum/pressure. (See 4.12).

3.12 Workmanship. The C-6709( )/G shall be manufactured and assembled in accordance with requirement 9 of MIL-STD-454 and the applicable portions of MIL-P-11268. (See 4.4).

3.13 Reliability. The C-6709( )/G shall demonstrate a specified mean-time-between-failure (MTBF) of 7000 hours, when tested in accordance with 4.11.

3.14 Electrical Performance of Assemblies (Modules) and Enclosure Assembly. Individual assemblies shall be tested to determine compliance with the performance requirement specified in 4.8.

3.15 Systems Safety Engineering.

3.15.1 Personnel Hazards. Personnel hazards shall be kept to a minimum. The criteria of MIL-STD-454, Requirement 1, shall be made applicable for tasks and efforts such as the selection of parts, the complete manufacturing and assembly process, and any product baseline configuration changes that may be implemented during the course of the contract. Verification that compliance with this requirement has been achieved shall be through a visual inspection. (See 4.13).

3.15.2 Edge Rounding Exposed. Exposed edges shall be rounded to a minimum radius of 0.04 inch (1 mm), and exposed corners to a minimum of 0.5 inch (13 mm).

3.15.3 Radioactive Materials. Radioactive materials shall not be used (e.g., luminous dials/markings, electron tubes, surge arrestors and lenses), other than those approved for use in the production drawings.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the government. The government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Test equipment and facilities for government inspections shall be made available by the contractor. The government reserves the right for authorized technical representatives to witness any contractor-performed inspections and tests.

4.2 Classification of Inspections. The inspection requirements specified herein are classified as follows:

a. First Article Inspection. (See 4.3). Does not include preparation for delivery.

b. Inspections covered by subsidiary documents. (See 4.4).

c. Quality Conformance Inspections.

(1) Quality conformance inspection of equipment before packaging(See 4.5).

(2) Quality conformance inspection of packaging. (See 4.15).

4.3 First Article. Unless otherwise specified in the contract or purchase order, the First Article inspection shall be performed by the contractor.

4.3.1 First Article Units. The contractor shall furnish six (C-6709( )/G) First Article units.

4.3.2 First Article Inspection. The First Article inspection shall consist of inspections specified in Table I and shall be performed in the order specified herein.

TABLE I

NOTES:

a. The inspections 1 through 5, in the order shown, shall be performed on all First Article units.

b. The implementation of inspections covered by 6 shall be expedited to satisfy contract delivery requirement.

c. The numbers in the unit columns in the table covered by 1-6 specify the order of inspections for the indicated units.

d. The equipment shall be thoroughly washed, cleaned, dried, and refurbished before proceeding with subsequent inspection.

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TABLE I. FIRST ARTICLE INSPECTION

Inspections	Rqmt Para	Test Para	Unit 1	Unit 2	Unit 3	Unit 4	Unit 5	Unit 6
1. Inspections Covered by Subsidiary Documents		4.4	Inspections to be performed on all units					
2. Bounce Preconditioning	3.7.1	4.6.1	Inspections to be performed on all units					
3. Electrical Preconditioning	3.7.2	4.6.2	"	"	"	"	"	
4. Group A Inspection	(See Table II)		"	"	"	"	"	
5. Group B Inspection	(See Table III)		"	"	"	"	"	
6. Group C Inspection a),b),c) SEQUENCE NO.								
1. Temperature	3.9.1	4.7.1	1					
2. Leakage (Immersion)	3.9.4	4.7.7		1/6				
3. Humidity	3.9.2	4.7.3	3					
4. Altitude	3.9.3	4.7.2	2					
5. Vibration	3.9.5	4.7.4.1		2				
6. Shock, Bench Handling	3.9.5.2	4.7.5		4				
7. Bounce, Loose Cargo	3.9.5.1	4.7.4.2		3				
8. Shock, Drop	3.9.5.3	4.7.6		5				
9. Dust (Fine Sand)d)	3.9.6	4.7.8	5	7				
10. Salt Fog d)	3.9.7	4.7.9				1		
11. Fungus d)	3.9.9	4.7.10					2	
12. Rain	3.9.8	4.7.11	4					
13. EMI	3.10	4.9					1	
7. Group D Reliability a),b),c)	3.13	4.11			1			1

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4.4 Inspections Covered by Subsidiary Documents. The following shall be inspected under the applicable subsidiary documents as part of the inspection required by this specification, and the inspection requirement specified in the contract or purchase order.

Item	Where Required
Construction	3.2
Parts, materials & processes general	3.3
Finish	3.4
Marking	3.5
Workmanship	3.12

4.5 Quality Conformance Inspection of Equipment Before Preparation for Delivery. The contractor shall perform the inspection specified in 4.4, 4.5.1 through 4.5.4 and 4.6. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product, and to assure compliance with all specification requirements.

4.5.1 Group A Inspection. Completed equipments shall be inspected for conformance to the inspections specified in Table II. Each assembly after 100% inspection, shall be formed into discrete lots for quality assurance verification. The equipments assembled from lots accepted under assembly verification shall be inspected 100% and formed into discrete lots for final QA verification. Factors of lot composition not defined herein, or in the contract or purchase order, shall be in accordance with MIL-STD-105. Each lot shall be submitted to sampling inspection, utilizing the procedures of MIL-STD-105 using general inspection levels and AQLs indicated in Table II (Quality Assurance Verification).

4.5.1.1 Order of Inspection Within Group A. Group A inspection shall be performed in an order satisfactory to the government, except that the operational test in 4.8.12 shall be next to last, and the air-seal test in 4.12 shall be last.

4.5.2 Group B Inspection. Group B inspection shall normally be performed on inspection lots that have passed Group A inspection and on samples selected from units that have been subjectd to and met the Group A inspection. This inspection shall conform to Table III and to the special inspection levels of Table I of MIL-STD-105.

4.5.2.1 Group B Sampling Plans. The Group B AQL shall be as shown in Table III and the inspection level shall be S-4.

4.5.2.2 Order of Inspection Within Group B. Within Group B inspection interchangeability (dimension) inspection shall be performed first, followed by the electrical tests in an order satisfactory to the government.

4.5.3 Group C Inspection. Group C isnpection shall be performed on units that have passed Group A and Group B inspection. The inspection shall consist of the inspection specified in Table IV. Samples shall be selected in accordance with 4.5.3.1.

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TABLE II

GROUP A Inspection

Insp	Reqmt	Insp	Major AQL	Minor
Visual-Mechanical	3.6	4.13	2.5%	4.0%
Workmanship	3.12	4.4	2.5%	4.0%
Finish	3.4	4.4		
Marking	3.5	4.4		
Electrical Performance				
Generated Signals	3.8.1	4.8.1		
Seize	3.8.1.1	4.8.1.1	1.0%	*
Release	3.8.1.2	4.8.1.2	1.0%	*
Acknowledge	3.8.1.3	4.8.1.3	1.0%	*
Ringback	3.8.1.4	4.8.1.4	1.0%	*
DTMF Signaling				
Dual Keys	3.8.1.5.1	4.8.1.5	1.0%	*
Detection	3.8.2	4.8.2		
Seize	3.8.2.1	4.8.2.1	1.0%	*
Release	3.8.2.2	4.8.2.2	1.0%	*
Seize/Release				
Acknowledge	3.8.2.3	4.8.2.3	1.0%	*
Control Signal(DTMF)	3.8.2.4	4.8.2.4	1.0%	*
Transmit On	3.8.2.4.1	4.8.2.4.1	1.0%	*
Transmit Off	3.8.2.4.2	4.8.2.4.2	1.0%	*
Control Signal(Single Freq)	3.8.2.5	4.8.2.5	1.0%	*
Transmit On	3.8.2.5.1	4.8.2.5.1	1.0%	*
Transmit Off	3.8.2.5.2	4.8.2.5.2	1.0%	*
Control Signal(AVOX)	3.8.2.6	4.8.2.6	1.0%	*
Audio Signals	3.8.6.1	4.8.6.1	1.0%	*
Operator's Handset/ Headset	3.8.7	4.8.7	1.0%	*
Acoustic Output (Headset)	3.8.7.1	4.8.7	1.0%	*
Sidetone (Headset)	3.8.7.2	4.8.7	1.0%	*
Handset Function(PTT)	3.8.7.3	4.8.7	1.0%	*
Operational Performance Test	3.8.12	4.8.12	1.0%	*
Air Seal Test	3.11	4.12	1.0%	*

*All electrical tests are major

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4.5.3.1 Sampling for Group C Inspection.

4.5.3.1.1 Subgroup 1. For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be selected from three month's production or 50 units, whichever is smaller.

4.5.3.1.2 Subgroup 2. For this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be selected from three-months' production or 75 units, whichever is smaller. Also for this subgroup, two units shall be selected at random from the first production lot. For subsequent Group C inspection, two units shall be selected from 4-months' production or 75 units, whichever is smaller.

TABLE III
GROUP B Inspection

Insp	Reqmt	Insp	AQL
Interchangeability	3.6	4.14	6.5%
Impedance Wireside	3.8.3.1	4.8.3.1	4.0%
Radio Side	3.8.3.2	4.8.3.2	4.0%
Receive Leg	3.8.3.2.1	4.8.3.2	4.0%
Transmit Leg	3.8.3.2.2	4.8.3.2	4.0%
Frequency Response	3.8.4	4.8.4	4.0%
Longitudinal Balance	3.8.5	4.8.5	4.0%
Amplitude Response	3.8.6	4.8.6	4.0%
Noise Level	3.8.8	4.8.8	4.0%
Power Requirements	3.8.9	4.8.9	4.0%
Polarity Protection	3.8.9.1	4.8.9.1	4.0%
Distortion	3.8.10	4.8.10	4.0%
Audible Alarm	3.8.11.1	4.8.11.1	4.0%
Visual Alarm	3.8.11.2	4.8.11.2	4.0%

4.5.3.2 Order of Inspection Within Group C. Group C inspection shall be performed in the order specified in Para 4.5.3.1.

4.5.3.3 Group C Failures. Actions required relative to Group C failures shall be as specified in the contract or purchase order.

4.5.4 Reinspection of Conforming Group B and Group C Sample Units. Unless otherwise specified, sample units which have been subjected to and passed both Group B and Group C inspections may be accepted on contract provided all damage is repaired and the sample units are resubjected to and pass Group A inspection.

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4.5.5 Group D Inspection. This inspection shall consist of the tests specified in Table V and shall be performed on units from lots which have been subjected to and met Group A and Group B inspection.

4.5.5.1 Sampling for Group D Inspection. For reliability purposes, two samples shall be selected at random once each three month's production, or 50 units whichever is smaller.

TABLE IV
GROUP C Inspection

Insp	Reqmt	Insp	Unit #1	Unit #2
Subgroup I				
Low Temperature	3.9.1.1	4.7.1.1	1	
High Temperature	4.9.1.2	4.7.1.2	2	
Immersion	3.9.4	4.7.7		1
Subgroup II				
Immersion	3.9.4	4.7.7	1 & 6	
Vibration	3.9.5	4.7.4.1	2	
Bounce	3.9.5.1	4.7.4.2	3	
Shock, Bench Handling	3.9.5.2	4.7.5	4	
Shock, Drop	3.9.5.3	4.7.6	5	
Humidity	3.9.2	4.7.3		1
Altitude	3.9.3	4.7.2	7	
Dust	3.9.6	4.7.8		2
Salt Fog*	3.9.7	4.7.9	8	
Fungus**	3.9.9	4.7.10		
Rain	3.9.8	4.7.11		3
Electromagnetic Compatibility	3.10	4.9	9	

* The equipment shall be thoroughly washed, cleaned, dried, and refurbished, if necessary, before proceeding with subsequent tests.

** This test performed on First Article unit only (see Table I).

TABLE V
GROUP D Inspection

Inspection	Requirements	Inspection
Reliability	3.13	4.11

4.5.5.2 Group D Failures. Actions required relative to Group D failures shall be as specified in the contract.

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4.5.5.3 Reinspection of Conforming Group D Sample Units. Unless otherwise specified, sample units which have been subjected to and passed group D inspection may be accepted on the contract or purchase order provided all visible and known damage is repaired and the sample units are resubjected to and pass Group A inspection.

4.6 Preconditioning (See 3.7).

4.6.1 Bounce Preconditioning. Each unit, in its nonoperating condition, shall be placed in its normal operating position on the table of the package tester, type 1000SC, as made by the L.A.B. Corporation, Skaneateles, NY or equal. The package tester, shafts in phase, shall have a speed such that it is just possible to insert a 1/32 inch thick strip of material under one corder or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. Each unit of each lot shall be subjected to this preconditioning. After bounce preconditioning, the unit shall not be repaired, aligned, cleaned or otherwise changed prior to subjection to acceptance inspection. (See 3.7.1).

4.6.2 Electrical Preconditioning (Burn-In). Each unit C-6709( )/G shall be subjected to the test conditions cited in 4.11.4 and shall meet the requirements of 3.7.2. At 24 hours in test, the operational performance test of paragraph 4.8.12 shall be performed.

4.7 Environmental Tests. The following tests shall be performed on the equipment. (See 3.9). When an electrical test is required, paragraph 4.8 shall be performed except as noted.

4.7.1 Temperature Tests (See 3.9.1).

4.7.1.1 Low Temperature. The equipment shall be subjected to the test of Method 502.1, Procedure I, of MIL-STD-810. The storage temperature (Step 2) shall be -70°F (-56.7°C) and shall be maintained for a period of not less than 2 hours following stabilization of the test item. Low operating temperature (step 4) shall be -40°F(-40.0°C). (See 3.9.1.1).

4.7.1.2 High Temperature. The equipment shall be subjected to the test of Method 501.1, Procedure II, of MIL-STD-810. For step 4, the chamber temperature shall be +160°F (+71°C). For step 7, the chamber temperature shall be adjusted to +150°F (-65°C) for test item operation (this includes effects of solar radiation). (See 3.9.1.2).

4.7.2 Altitude Test (See 3.9.3). The equipment shall be tested in accordance with Method 500.1, Procedure I of MIL-STD-810. For step 2, the chamber altitude shall be 15,000 feet above sea level. In addition, the equipment shall withstand air transportation altitude up to 40,000 feet above sea level while nonoperating. Only the operational test (4.8.12) shall be performed.

4.7.3 Humidity Test (See 3.9.2). The equipment shall be subjected to test Method 507.1, Procedure III of MIL-STD-810. In step 6, measurements shall be taken during the last five (5) hours of the last cycle.

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4.7.4.1 Vibration (See 3.9.5). The equipment shall be subjected to the test of Method 514.2, Procedure X. Figure 5.4.2-6 Curve AW of MIL-STD-810. The time schedules shall be in accordance with Table 5.14.2-VII of MIL-STD-810. The equipment shall be electrically operated and mechanically inspected before and after test and the two results compared.

4.7.4.2 Bounce, Loose Cargo (See 3.9.5.1). The equipment in the field portable (loose cargo) configuration shall be subjected to Test Method 514.2, Procedure XI, Part 2 of MIL-STD-810.

4.7.5 Shock Test - Bench Handling (See 3.9.5.2). The equipment shall be subjected to Test Method 516.2, Procedure V of MIL-STD-810.

4.7.6 Shock Test Drop (See 3.9.5.3). The equipment shall be subjected to Test Method 516.2, Procedure II of MIL-STD-810.

4.7.7 Immersion Test (See 3.9.4). The equipment shall be subjected to Test Method 512.1, Procedure I of MIL-STD-810. This test shall be performed prior to and following vibration and shock testing of the same test item.

4.7.8 Dust Test (See 3.9.6). The equipment shall be subjected to Test Method 512.1, Procedure I of MIL-STD-810.

4.7.9 Salt Fog Test (See 3.9.7). The equipment shall be subjected to Test Method 509.1, Procedure I of MIL-STD-810. Operation of the equipment shall not be required and failure criteria shall be limited to corrosion of finishes and metals only. Such corrosion shall be defined as any visible degradation of the equipment surfaces that can be attributed to flaky, pitted blistered or otherwise loosened finish on metal surface.

4.7.10 Fungus Test (See 3.9.9). The equipment shall be subjected to Test Method 508.1, Procedure I, of MIL-STD-810. The equipment shall be opened and the parts and materials of the interior circuit assemblies shall be sprayed with spore suspension. The interior and exterior of the case shall also be sprayed. Operation is not required. Inability of the equipment to meet the requirements of 3.9.9 shall constitute failure of the test.

4.7.11 Rain. The equipment shall be subjected to Test Method 506.1, Procedure I of MIL-STD-810. The equipment shall be operated during the last 10 minutes of each 30 minute period. (See 3.9.8).

4.8 Electrical.

4.8.1 Generated Signals

4.8.1.1 Seize (See 3.8.1.1). Set up the circuit of Figure 14b. Press oper connect SWBD button on the C-6709( )/G. The VTVM 2 level shall read -7 dBm ± 2 dB and EC shall read 2250 Hz ± 29.3 Hz.

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4.8.1.2 Release (See 3.8.1.2). Set up the circuit of Figure 14b. Press Release button on the C-6709( )/G. The VTVM 2 level shall read -7 dBm ± 2 dB and EC shall read 2600 Hz ± 33.8 Hz. Time out as measured on the scope shall be 2 to 10 seconds.

4.8.1.3 Seize/Release Acknowledge (See 3.8.1.3). Set up the circuit of Figure 14b. Adjust the OSC frequency to 2250 Hz ± 29.3 and output level to .31 Vrms as read on VTVM 1. Move SW-1 to position A. The VTVM 2 level shall read -7 dBm ± 2 dB and EC shall read 570 Hz ± 7.4 Hz. Note on scope that 570 Hz ± 7.4 Hz shall be cut off after 250 ms upon cessation of seize signal. Repeat preceding procedure for release signal 2600 Hz ± 33.8 Hz. Be sure the unit is in a busy condition before performing the release procedure.

4.8.1.4 Ringback (See 3.8.1.4). Set up the circuit of Figure 14b and set SW-1 to positon A. Inhibit the 425 Hz interrupter. Move SW-1 to position B. The VTVM 2 shall read -7 dBm ± 2 dB and EC shall read 425 Hz ± 5 Hz. Remove the 425 Hz interrupter inhibit and observe ringback waveform on scope A 425 Hz, waveform shall be on for 2 sec ± 400 ms and off for 4 sec ± 800 ms. While on, the 425 Hz shall be modulated 50 ms on the 50 ms off ± 10 ms.

4.8.1.5 DTMF Signaling. Set up circuit as shown in Figure 14b. Test the DTMF signaling to insure compliance with 3.8.1.5.1.

4.8.2 Detection

4.8.2.1 Seize (See 3.8.2.1.). Set up the test circuit of Figure 14a with the attenuator to 0 dB. Disconnect oscillator #1 and set oscillator #2 to 2250 Hz nominal frequency at an output of 0 dBm as read on the VTVM. Set the attenuator to maximum. Move the switch to the up position. Slowly decrease attenuator setting until the alarm and the status lamp activate. The seize signal 2250 Hz shall be detected with attenuator setting between 5 dB to 30 dB. Repeat the above procedure for oscillator frequencies of 2205 Hz and 2295 Hz. Using a tone burst gate, determine and record the time necessary to detect each of the three frequencies. Detection time shall be 700 ms ± 100 ms. Set attenuator at +5 dB and OSC at 2139.5 Hz, seize shall not be detected. Next set OSC at 2362.5 Hz, seize shall not be detected.

4.8.2.2 Release (See 3.8.2.2). Repeat the test procedure as indicated in section 4.8.2.1 with oscillator settings of 2548 Hz, 2652 Hz and 2600 Hz nominal frequency. Be sure the unit is in a busy condition before performing this procedure. Release signal has been detected at the time when a 570 Hz signal appears on the send terminals. Set atten at 5 dB and OSC at 2756 Hz, release shall not be detected. Next set OSC at 2444 Hz, release shall not be detected. Repeat above procedure with the unit in the idle condition.

4.8.2.3 Seize/Release Acknowledge (See 3.8.2.3). Repeat the test procedure as indicated in section 4.8.2.1 with oscillator setting of 542 Hz, 498 Hz, and 570 Hz nominal frequency. The detection time shall be 450 ms + 50 ms. Set atten at 5 dB and OSC at 541 Hz, seize/release acknowledge shall not be detected. Next set OSC for 598 Hz, seize/release acknowledge shall not be detected.

4.8.2.4 Control Signal DTMF

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4.8.2.4.1 Transmit On (See 3.8.2.4.1). Set up the test circuit of Figure 14a with the attenuator to 0 dB. Adjust oscillator #1 to 697 Hz at an output of 0 dB as read on VTVM with oscillator #2 disconnected. Adjust oscillator #2 to 1209 Hz at an output of 0 dB as read on VTVM with oscillator #1 disconnected. Connect both oscillators back into the circuit. Set the attenuator to maximum and switch to up position. Slowly decrease attenuator setting until the transmitter activates. The signal shall be detected with attenuator setting between 5 dB to 30 dB. Detection time of 150 ms ± 25 ms shall be measured by using a tone burst gate. Adjust oscillator #1 to 711 Hz and oscillator #2 to 1233 Hz. Repeat the above procedure. Adjust oscillator #1 to 683 Hz and oscillator #2 to 1185 Hz. Repeat the above procedure. Set OSC #1 for 662 Hz, transmit shall not be turned on. Next set OSC for 731 Hz, transmit shall not be turned on.

4.8.2.4.2 Transmit Off (See 3.8.2.4.2). Operate the switch to down position, change oscillator #2 frequency to 1477 Hz. Operate the test switch to the up position for several seconds and back to the down position. When the switch is returned to the down position, the transmitter will be switched off by the C-6709()/G with the attenuator settings between 5 dB to 30 dB. Detection time of 150 ms ± 25 ms shall be measured by using a tone burst gate. Adjust oscillator #2 to 1506 Hz and repeat the above procedure. Adjust oscillator #2 to 1448 Hz and repeat the above procedure.

4.8.2.5 Control Signal (Single Frequency)

4.8.2.5.1 Transmit On (See 3.8.2.5.1). Repeat the test procedure as indicated on section 4.8.2.4.1 with oscillator #1 disconnected and oscillator #2 frequency of 1209 Hz ± 24.2 Hz. The detection time shall be 375 ms ± 50 ms. Set OSC #2 for 1148 Hz, transmit shall not turn on. Next set OSC #2 for 1269 Hz, transmit shall not turn on.

4.8.2.5.2 Transmit Off (See 3.8.2.5.2). Repeat the test procedure as indicated on section 4.8.2.4.2. with oscillator #1 disconnected and oscillator #2 frequency of 1477 Hz ± 29.6 Hz. The detection time shall be 375 ms ± 50 ms. Set OSC #2 for 1403 Hz, unit shall not detect the signal. Next set OSC #2 for 1550 Hz, unit shall not detect the signal.

4.8.2.6 Control Signal (AVOX) (See 3.8.2.6). Move the C-6709( )/G mode switch to AVOX. The transmit keying shall be activated (LED on) between 100 ms and 200 ms from the beginning of speech and the transmit unkeying shall be activated (LED off) between 500 ms and 1000 ms from ending of speech. Record the average time for both the transmit keying and transmit unkeying. Set up circuit of Figure 13. Adjust noise generator output to -25 dBm and attenuator setting to maximum. Decrease attenuator gradually until transmit keying occurs and note this level as read on VTVM. Repeat and record the average level at which transmit keying occurs. Repeat preceding measurement without the noise generator.

4.8.3 Impedance

4.8.3.1 Wire Side (See 3.8.3.1). Set up the circuit of Figure 4, measuring the magnitude and phase angle of the impedance of the receive terminals on the switchboard side of the C-6709( )/G. With the DECADE/UUT switch in the UUT position, adjust the oscillator frequency to 300 Hz, and an output level of -4 dBm as read

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on the VTVM. Adjust the decade resistor box such that, when the DECADE/UUT switch is in either the DECADE or UUT position, the VTVM shows an identical reading. The value of the decade resistor box setting is the value of the absolute magnitude of the impedance looking into the C-6709( )/G. Next adjust the oscilloscope so that the amplitudes of trace 1 and 2 appear identical. Switch the oscilloscope to the most sensitive reading maintaining the traces at the same amplitude. Note the time difference (t) in the zero crossing between the two traces, the phase angle $\theta=360$ (t) (f), where f is the frequency of the oscillator. Repeat the above procedure with the oscillator set to test frequencies of 500 Hz, 1000 Hz, 2000 Hz, 4000 Hz, 10,000 Hz, 30,000 Hz 50,000 Hz and 70,000 Hz. Set up the circuit of Figure 4a and measure the magnitude and phase of the send terminals on the switchboard side of the C-6709( )/G. Follow the same procedure above for the receive terminals.

4.8.3.2 Radio Side (See 3.8.3.2). Set up the circuit of Figure 6. With the DECADE/UUT switch in the UUT position, adjust the oscillator frequency for 300 Hz and an output of .15 Vrms as read on the VTVM. Adjust the decade resistor box such that, when the DECADE/UUT switch is in either the DECADE or UUT position, the VTVM shows an identical reading. The value of the decade resistor box setting is the value of the absolute magnitude of the impedance looking into the C-6709( )/G. Next adjust the oscillator so that the amplitude of trace 1 and 2 appear identical. Switch the oscilloscope to the most sensitive reading, maintaining the traces at the same amplitude. Note the time difference (t) in the zero crossing between the two traces, the phase angle $\theta=360$ (t) (f), where f is the frequency of the oscillator. Repeat the above procedure with the oscillator set to test frequencies of 500 Hz, 1,000 Hz, 2,000 Hz, 4,000 Hz, 10,000 Hz 30,000, 50,000 Hz and 70,000 Hz. Set up the circuit of Figure 5. Measure the absolute magnitude of the impedance looking into the send terminals, radio side of the C-6709( )/G. Adjust the oscillator for 300 Hz and with the decade resistor set for zero ohms, adjust the input signal level for 1 mVrms across the 10 ohm resistor. Verify that the signal is not clipped. Adjust the decade resistor until the voltage across the 10 ohm resistor is .99 mVrms. The magnitude is then determined by the formula: Magnitude + 99X decade resistor value. Repeat the above procedure with the oscillator set to test frequencies of 500 Hz, 1,000 Hz, 2,000 Hz 4,000 Hz, 10,000 Hz, 30,000 Hz and 70,000 Hz.

4.8.4 Frequency Response (See 3.8.4). Set up the circuit of Figure 7 and measure the frequency response of the C-6709( )/G from the wire side to the radio side. Adjust the oscillator frequency to 1 KHz at an output level of -16 dBm as read on the VTVM. Maintain this level throughout the test. Set the Radio Level switch to position 1. In positions 1, 2 and 3 a 600 ohm termination is required across the radio send terminals. In positions 4, 5, 6, 7 and 8 a 150 ohm termination is required. Measure the output level at 1 KHz, 300 Hz, 500 Hz, 2 KHz, 30 KHz, 50 KHz and 70 KHz. Repeat the measurements for RADIO LEVEL positions 2, 3, 4, 5, 6, 7 and 8. For each measurement the deviation from 1 KHz is calculated by dB=20 log $\frac{\text{Vrms measured}}{\text{Vrms at 1 KHz}}$. Next set up the circuit of Figure 8, for measuring the frequency response from the radio side of the C-6709( )/G to the switchboard side. Adjust the oscillator for 1 KHz and a level of .22 Vrms as read on the VTVM. Maintain this level throughout the test. Measure the output level as read on the WA at 1 KHz, 300 Hz, 500 Hz, 2 KHz, 10 KHz, 30 KHz, 50 KHz, and 70 KHz. Calculate the deviation from 1 KHz using the above formula.

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4.8.5 Longitudinal Balance (See 3.8.5). Set up the circuit of Figure 9 measuring the transverse voltage of the receive terminals on the wire side of the C-6709( )/G. Place the C-6709( )/G RADIO LEVEL switch in position 4. Adjust the level at the radio side. This level is the transverse voltage. Repeat for frequencies of 500 Hz, 1 KHz, 2 KHz, 10 KHz, 30 KHz, 50 KHz, 70 KHz. Next, set up the circuit of Figure 4A measuring the longitudinal voltage of the receive terminals on the wire side of .49 Vrms. Measure the level at the radio side. This level is the longitudinal voltage. Repeat for frequencies of 500 Hz, 1 KHz, 3 KHz, 4 KHz, 10 KHz, 30 KHz, 50 KHz and 70 KHz. Calculate the longitudinal balance for each frequency in dB using the following equation:

$$\text{Long. Bal in dB} + 20 \log \frac{\text{Longitudinal Voltage}}{\text{Transverse Voltage}}$$

4.8.6 Amplitude Response (See 3.8.6). Set up the circuit of Figure 7 to measure the audio level amplitude response between the wire receive terminals and the transmitter terminals. Place the C-6709( )/G RADIO LEVEL switch in position 1. Adjust the oscillator to a frequency of 1 KHz and a level of .25V as read on the VTVM. Measure the level at the radio port as read on the Wave Analyzer (WA). Repeat with the oscillator level set for .2, .15, .1, .05 and .15 volts. Repeat the above procedure for switch positions 2 through 8. Plot a curve of wire terminals receive signals in volts rms vs radio terminals send signals in millivolts rms. The curve shall be approximately linear.

4.8.6.1 Audio Signals (See 3.8.6.1). Set up the circuit as shown on Figure 8 and the C-6709( )/G RADIO LEVEL switch in position 1. Adjust the oscillator to a frequency of 1 KHz and an output of 0.22V as read on the VTVM. Measure the level set for 0.1V and 0.05V as read on the VTVM. Plot a curve of radio terminals received signal in volts rms vs wire terminals send signal in volts rms. The curve shall be approximately linear.

4.8.7 Operator's Handset/Headset (See 3.8.7.1, 3.8.7.2, 3.8.7.3). Set up the test circuit of Figure 14C. Set the oscillator to 2250 Hz and adjust the oscillator level until a service request condition occurs in the C-6709( )/G. Remove the oscillator. Momentarily operate the OPER CONNECT SWBD pushbutton. The audible alarm extinguishes and the SVC REQ/BUSY lamp illuminates steadily. Connect the OSC to the transformer as shown in the figure. Place the sound level meter 1/16" in front of the TA-235 speaker. Adjust the OSC level until the meter reads 93 dB at 1 KHz. Next, place the TA-235 speaker 1/16' in front of the headset microphone. Place the sound level meter 1/16' in front of the headset earcup and measure the sound pressure level. Measure also the audio level at the 4-wire switchboard send port. The VTVM shall read -20 dBm minimum. Since the sound level at the microphone is 93 dB above .0002 Dynes/CM^2, then the level at the earcup (sidetone), shall be 20 $\pm$ 4 dB below that or 73 $\pm$ 4 dB above .002 Dynes/CM^2. Connect the H-250 handset to the headset connector on the C-6709( )/G. Depress the push-to-talk button and verify that the C-6709 operator can activate XMIT keying as long as he is connected to the radio side.

4.8.8 Noise Level (See 3.8.8). Set up the circuit of Figure 10, measuring the level of noise generated by the C-6709( )/G. Place the C-6709( )/G RADIO LEVEL switch in position 4 and set up a call with the C-6709( )/G in the transmit condition. Place the noise measuring set in the F1A position. With the test switch

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in the up position, measure the noise generated at the radio side port. Next, place the switch in the down position and measure the noise generated at the wire side port.

4.8.9 Power Requirements (See 3.8.9). Connect C-6709( )/G as shown on Figure 4. Adjust power supply voltage to 24 volts ± 5% Connect + lead of DVM to C-6709( )/G power switch terminal and - lead of DVM to negative power binding post to C-6709 ( )/G. Operate C-6709( )/G power switch to internal. DVM shall read 24 volts ± 5%. Remove fuse from C-6709( )/G. DVM shall read 0 volts. Replace fuse. DVM shall read 24 volts ± 5%. Next, restore C-6709( )/G power switch to off. DVM shall read 0 volts. Operate C-6709( )/G power switch to external. DVM shall read 24 volts ±5%. Remove fuse from C-6709( )/G. DVM shall read 0 volts. Replace fuse DVM shall read 24 volts ± 5%. Next connect DVM + lead to pin 13 on PWB connectors. DVM shall read + 15 VDC ± 5%. Connect DVM + lead to pin 12 on PWB connectors. DVM shall read +7.5 VDC ± 5%. Next, adjust power supply voltage to +22 VDC ± 5% DVM shall read +15 VDC ± 5% on pin 13 of PWB connectors. DVM shall read ± 7.5 VDC ± 5% on pin 12 of PWB connectors. Next, adjust external power supply voltage to +32 VDC -5%, +0%. DVM shall read +15 VDC ± 5% on pin 15 of PWB connectors. DVM shall read +7.5 VDC ±5% on pin 12 of PWB connectors. Next, connect ammeter in series with external power supply leads to C-6709( )/G. Verify that ammeter reading is less than 200 milliampers at power supply input voltage of +24 VDC when the busy lamp is illuminated.

4.8.9.1 Polarity Protection (See 3.8.9.1). Reverse polarity. Reverse external power supply lead to C-6709( )/G. Verify that DVM reading is 0 volts and that C-6709( )/G fuse is not blown. Restore external power supply leads to normal.

4.8.10 Distortion (See 3.8.10). Set up the circuit of Figure 11, to measure the harmonic distortion of the C-6709( )/G Radio Control Set. Place the C-6709( )/G RADIO LEVEL switch in position 4. Set up the C-6709)/G so that it is in the transmit condition. Adjust the oscillator to a frequency of 300 Hz. Adjust the wave analyzer for a maximum reading at 300 Hz. Adjust the level of the oscillator for a reading of 8 mv as read on the wave analyzer. Adjust the analyzer frequency to the second harmonic of the oscillator. Record the level as read on the wave analyzer. Adjust the wave analyzer to the third harmonic of the oscillator. Record the level as read on the wave analyzer. Adjust the wave analyzer frequency to the fourth harmonic of the oscillator. Record the level as read on the wave analyzer. Repeat the above for an oscillator frequency of 1 KHz and 3.5 KHz. Any harmonic level shall not exceed .08 mv. Next, set up the circuit of Figure 12. Adjust the oscillator to a frequency of 300 Hz. Adjust the wave analyzer for maximum reading at 300 Hz and the oscillator level of 300 mv. Adjust the wave analyzer frequency to the second harmonic of the oscillator and record the level as read on the analyzer. Adjust the wave analyzer for the third harmonic of the oscillator and record the level as read on the wave analyzer. Adjust the wave analyzer frequency to the fourth harmonic of the oscillator and record the level as read on the wave analyzer. Repeat the above procedure for an oscillator frequency of 1.0 and 3.5 KHz. Any harmonic level shall not exceed 3 mv.

4.8.11 Audible and Visual Alarms (See 3.8.11).

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4.8.11.1 Audible Alarm (See 3.8.11.1). Momentarily operate the RELEASE pushbutton. Set up the test of Figure 14C. Set the OSC to 2250 Hz and adjust the OSC level until a service request condition occures in the C-6709( )/G. Measure the loudness of the alarm of the C-6709( )/G with the volume control to full clockwise. Measurement should be taken at a distance of 3 feet. Verify that the status lamp is flashing.

4.8.11.2 Visual Alarm (See 3.8.11.2). Momentarily operate the RELEASE pushbutton. Set up the test of Figure 14C. Set the OSC to 2250 Hz and adjust the OSC level until a service request condition occurs at the C-6709( )/G. Disconnect the oscillator and depress the OPER CONNECT SWBD pushbutton. Verify that the busy indicator remains on.

4.8.12 Operational Tests (See 3.8.12). With circuit setup as shown in Figure 5A, perform the test as indicated below.

4.8.12.1 Call to C-6709( )/G. Turn on supply and set for 24 Vdc then turn off. Set multimeter to monitor dc voltages up to and including 24 Vdc. Set the CALL ALARM and the HEADSET OFF-LOUD controls to mid-range. Set the TA-341/TT telephone, PH-1, to AC SUPV and LB and set TA-341/TT telephone PH-2 to PT-to-PT and LB. Place the C-6709( )/G Mode Switch in OPER and EXT-OFF-INT switch to EXT. Set RADIO LEVEL switch to position 1. Turn on power supply and check for 24 VDC. Lift PH-1 Handset. Verify that the SVC REQ/BUSY lamp flashes, the audible alarm sounds and ringback is heard on PH-1. Momentarily press the OPER CONNECT SWBD pushbutton and verify that the alarm is silent and the BUSY lamp is steady. Verify the talk path between the operator and telephone PH-1. Momentarily press the OPER CONNECT RADIO pushbutton and the digit 1 on the C-6709( )/G keypad. Verify that telephone PH-2 can hear the operator and the meter indicated 0 Vdc. Momentarily press digit 3 on the C-6709( )/G keypad. Verify that the operator can hear telephone PH-2 and the meter indicates 24 Vdc or greater. Momentarily press CALL CONNECT OPER IN pushbutton. Momentarily press digit 1 on the C-6709( )/G keypad. Verify that PH-2 can hear audio from the operator and PH-1, and the meter reads 0 Vdc. Rotate RADIO LEVEL switch through each position and make sure that the talk path between PH-1 and PH-2 is complete. Signal level will vary with switch position. Set switch to position 3. Momentarily press digit 3 on the C-6709( )/G keypad. Verify the talk path between the operator and PH-1, and that the operator and PH-1 can hear PH-2. The meter should read 24 Vdc. Momentarily press the CALL CONNECT OPER OUT pushbutton. Verify there is no talk path between the operator and PH-1, and the operator cannot hear PH-2. Place the C-6709( )/G mode switch in the DTMF position. Press digit 1 on PH-1 and hold for 1/2 second. The meter should indicate 0 Vdc when the digit 1 pushbutton is released. Press digit 3 on PH-1 and hold for 1/2 second. The meter should indicate about 24 Vdc when the digit 3 pushbutton is released. Place the C-6709( )/G mode switch in the SF(PTT) position. Press digits 1 and 4 simultaneously and hold for 1/2 second. The meter should indicate 0 Vdc. Press digits 3 and 6 simultaneously and hold for 1/2 second. The meter should indicate 24 Vdc. Place the C-6709( )/G mode switch in the AVOX position. Talking into PH-1 will cause the meter to read 0 Vdc for the duration of voice signals from PH-1. When the voice signals from PH-1 cease, the meter will increase to 24 Vdc after a delay of 1 second. Listen on PH-1 and press each key on the C-6709( )/G keypad. Verify presence of dual tone frequencies. Place PH-1 handset on hook and verify that SVC REQ/BUSY lamp extinguishes, indicating release.

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4.8.12.2 Call from C-6709( )/G. Set TA-341/TT PH-1, to PT-to-PT and LB. Momentarily press the OPER CONNECT SWBD pushbutton. Wait approximately 1 sec and lift handset from PH-1. Verify that the talk path between the operator and telephone PH-1 is established and that the SVC REQ/BUSY lamp is on steady. Momentarily press the RLSE pushbutton. A release signal should be heard in PH-1's handset. The release signal should time out between 3 and 10 seconds. Verify that the SVC REQ/BUSY lamp extinguishes when release signal times out.

4.9 Electromagnetic Interference (See 3.10).

4.9.1 Electromagnetic Compatibility Test (See 3.10.1; 3.10.2. The C-6709( )/G shall be tested for compliance with the requirements of 3.10.1. Test setups and test procedures shall conform to the measurement techniques of MIL-STD-462, Notice 3. Signal and data input and output terminals shall be terminated with simulated inputs and loads. Simulated loads shall have characteristic impedance similar to actual loads. All cables used during test shall be of the same type and length to be supplied by the contractor or used in the actual installation.

4.10 Bonding Impedance Test (See 3.10.3). Prior to the performance of the EMI/EMC testing of 4.9.1 bonding impedances shall be measured for compliance with the requirements of 3.10.1. All measurement readings shall be recorded and included in the EMI/EMC test report. Measurements shall be performed from the equipment case to the shell of case-mounted connectors to all other externally-mounted connectors. Additionally, measurements shall be performed between all mating surfaces bonded by EMI gasketing.

4.11 Reliability (See 3.13).

4.11.1 First Article Reliability Length/Accept/Reject Criteria. The reliability of the equipment shall be demonstrated by selecting two (2) First Article Units C-6709( )/G that have passed Group A and Group B inspection and subjecting them to a total of 1295 operating hours. The accept/reject criteria Are:

Accept on 0 failures
Reject on one (1) or more failures

4.11.2 First Article Reliability Test. The First Article reliability inspection testing shall be in accordance with Test Plan XXV of MIL-STD-781B.

4.11.3 Quality Conformance (Group D) Reliability Test. The reliability test for Group D inspection shall be in accordance with Test Plan XXV of MIL-STD-781B. The equipment reliability requirements of 7000 hours MTBF shall be demonstrated by completing a fixed length test time of 1295 hours without a failure. Two test units, each accumulating 24 hours on-time per day shall be used throughout the test period of 54 days.

4.11.4 Reliability Test Conditions. The reliability test shall be conducted at Test Level C, paragraph 4.1.4 of MIL-STD-781B. The equipment on-off cycling and duty cycle shall be as follows:

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a. Equipment turned on.

b. 24 calls shall be placed, each call consisting of 10 receive mode periods, and 10 transmit mode periods.

c. Equipment turned off.

Four of the above cycles shall be run per day. The remaining time, the equipment shall be in the standby mode. Proper instrumentation shall be provided to insure recognition of failure. A failure is defined as the inability of an equipment to perform its required function. This includes catastrophic failures as well as any event which causes any departure from performance as required by this specification or the contractor's approved test procedure. Paragraphs 5.5 through 5.9 inclusive of MIL-STD-781B shall apply.

4.11.5 Rework and Retest Provision. In case of failure to meet reliability test requirements, corrective action related to the observed failures shall be incorporated in all equipments of the lot before a new random sample is selected for reliability acceptance retest. Any subsequent rerun must start at zero hours operating time.

4.12 Air-Seal Test (See 3.11). The C-6709( )/G shall be opened and closed again in such a manner as to break and remake the seal. The air equalization vent shall be locked by suitable means. Immediately thereafter, the equipment as field-transported shall be subjected to a vacuum of one (1) pound per square inch (one pound per square inch less than the atmospheric pressure surrounding the equipment) applied to the interior of the equipment enclosure. The vacuum then shall be valved off, and the interior pressure measured during the ensuing period of one (1) minute. During this one-minute period the decrease in vacuum shall not exceed 0.01 pound. The guage used for measurement of the vacuum shall be of such accuracy that a difference of 0.01 pound can be determined readily.

4.13 Visual and Mechanical Inspection (See 3.6 and 4.14). The equipment shall be examined for all defects listed in MIL-STD-252. An inspection shall be performed to verify compliance with those portions of systems safety engineering (3.15) which can be determined visually.

4.14 Inspection for Dimensional Interchangeability. The dimensions listed below shall be gaged or measured to determine conformance to the physical interchangeability requirements of 3.5. When a listed dimension is not within specified or design limits, it shall be considered a major defect.

a. External and internal dimensions of cases, covers and insertable assemblies, when such dimensions affect mating of parts.

b. Dimensions of cavities, when such dimensions affect insertion of items.

c. Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

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d. Location of connectors, locking pins, fasteners, slides and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

e. Size and form of special threads.

4.15 Quality Conformance Inspection for Packaging. Quality conformance inspection for packaging shall be in accordance with Group A, Group B and, when required (see 6.2c), the Group C (rough handling) requirements of MIL-P-116. Inspection lots shall be in accordance with MIL-P-116.

5. PACKAGING

5.1 Packaging Requirements. The requirements for packaging shall be in accordance with SPI 1G0062.

6. NOTES

6.1 Intended Use. Control, Radio Set C-6709( )/G is intended to provide radio/wire integration facilities for telephone subscribers in switched communication systems. When connected between an analog or hybrid switchboard and a single channel (half-duplex) push-to-talk radio set the equipment shall provide the capability for answering calls from and for extending calls to DTMF, ring-down and digital telephones and for controlling the radio transmitter keying.

6.2 Ordering Data. Procurement documents should specify the following:

a. Title, number, and date of this specification and any amendment thereto.

b. Level A or Level B Preservation and Packing (see Section 5).

c. When rough handling test is required (see 4.15).

d. When First Article samples are required (see 4.3).

6.3 Environmental. Environmental pollution prevention measures are contained in the packaging material specifications referenced herein. Refer to material specifications or preparing activity for recommended disposability methods.

6.4 Group C Inspection. Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action (See 4.5.3.3).

6.5 Nomenclature. The parentheses in the nomenclature will be deleted or replaced by a letter identifying the particular design; for example: C-6709( )/G. The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.6 Verification Inspection. Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in the specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractors quality control system and the quality history of the product.

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6.7 Location of Operational Inspection. It is desirable that the operational inspection (4.8.12) be performed at a location that will minimize handling (which might cause damage to the equipment) after this inspection is complete. It is recommended that the entire lot (including all previously inspected sample units) be sampled and inspected immediately prior to packaging.

6.8 Definitions

6.8.1 Branching. Branching is a connected arrangement of filaments (hyphol) formed by shoots or secondary items growing from the main stem or filament(hypha).

6.8.2 Growth Colonization. Growth colonization is a mass of individual plants, generally of one species, living together; or a group of hyphae which is formed from one spore or cell and may be one individual plant. Colonization which completely covers the surface of the nutrient material constitutes abundant growth.

6.8.3 Sporulation. Sporulation is the formation of minute unicellular reproductive or dormant bodies, called spores.

6.8.4 Tubular Germination. Tubular germination is partial growth by the production of hypha, which are tubular shaped fungae filaments. Tubular germination constitutes restricted individual spore growth not proceeding to colonization.

6.8.5 Microbial Growth. Microbial growth is the growth of a very minute organism. Such organisms, when present in large numbers, may provide a colonly visible to the naked eye.

6.8.6 Exfoliation. Exfoliation is corrosion along the grain boundaries of the metal resulting in the peeling or separating, or both, of successive layers of the metal. The appearance resembles loose book pages or onion skin peeling.

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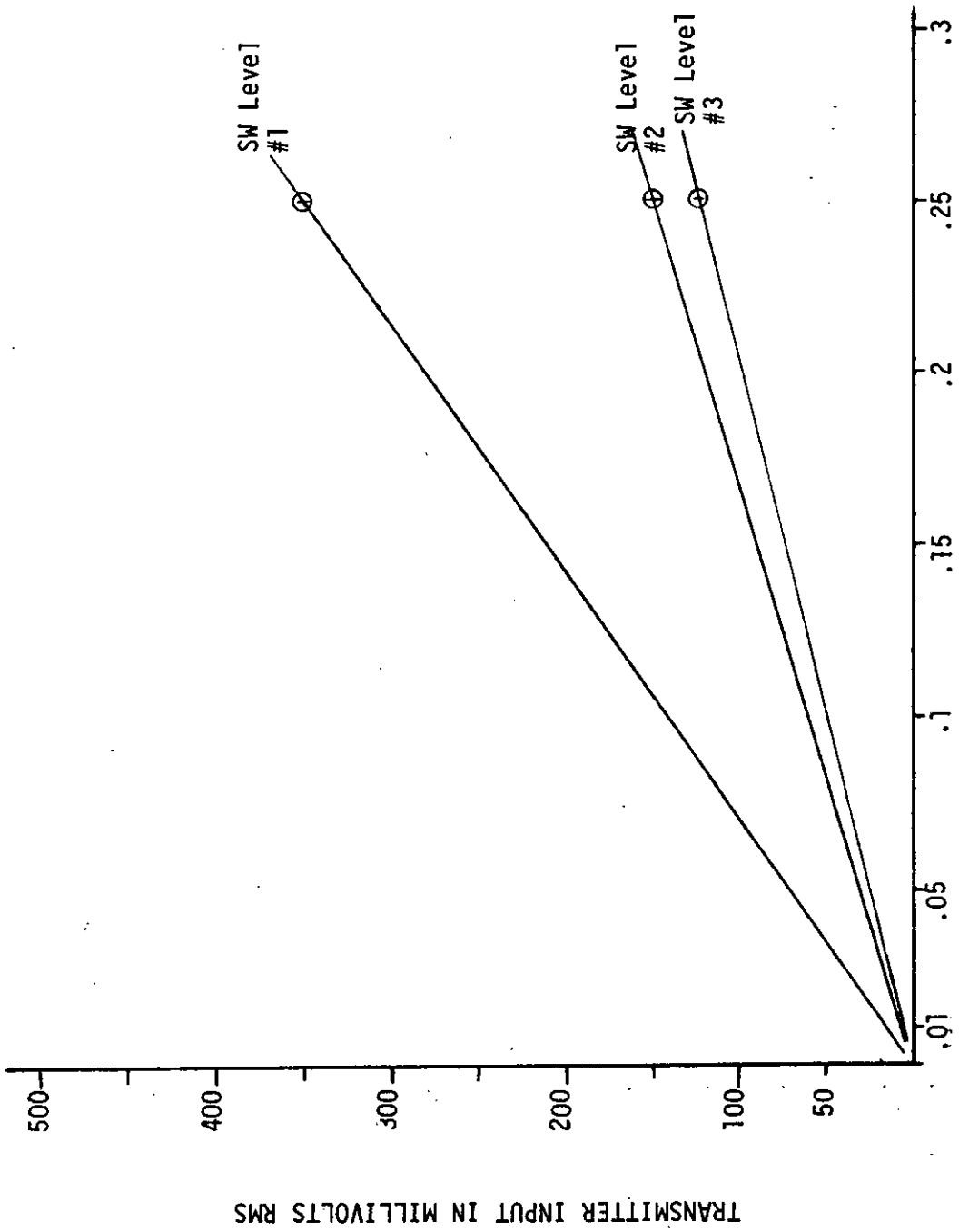


FIGURE 1

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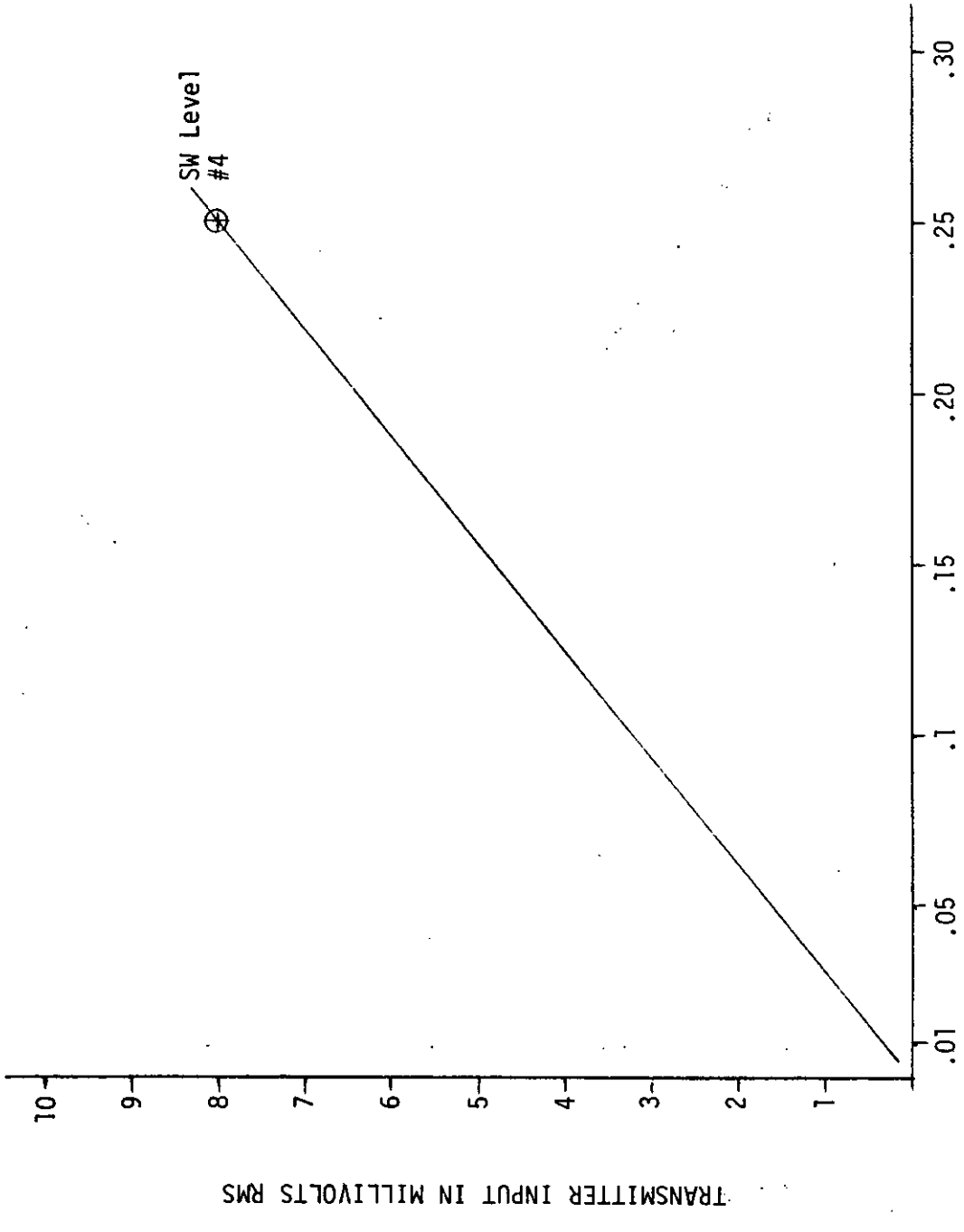


FIGURE 2

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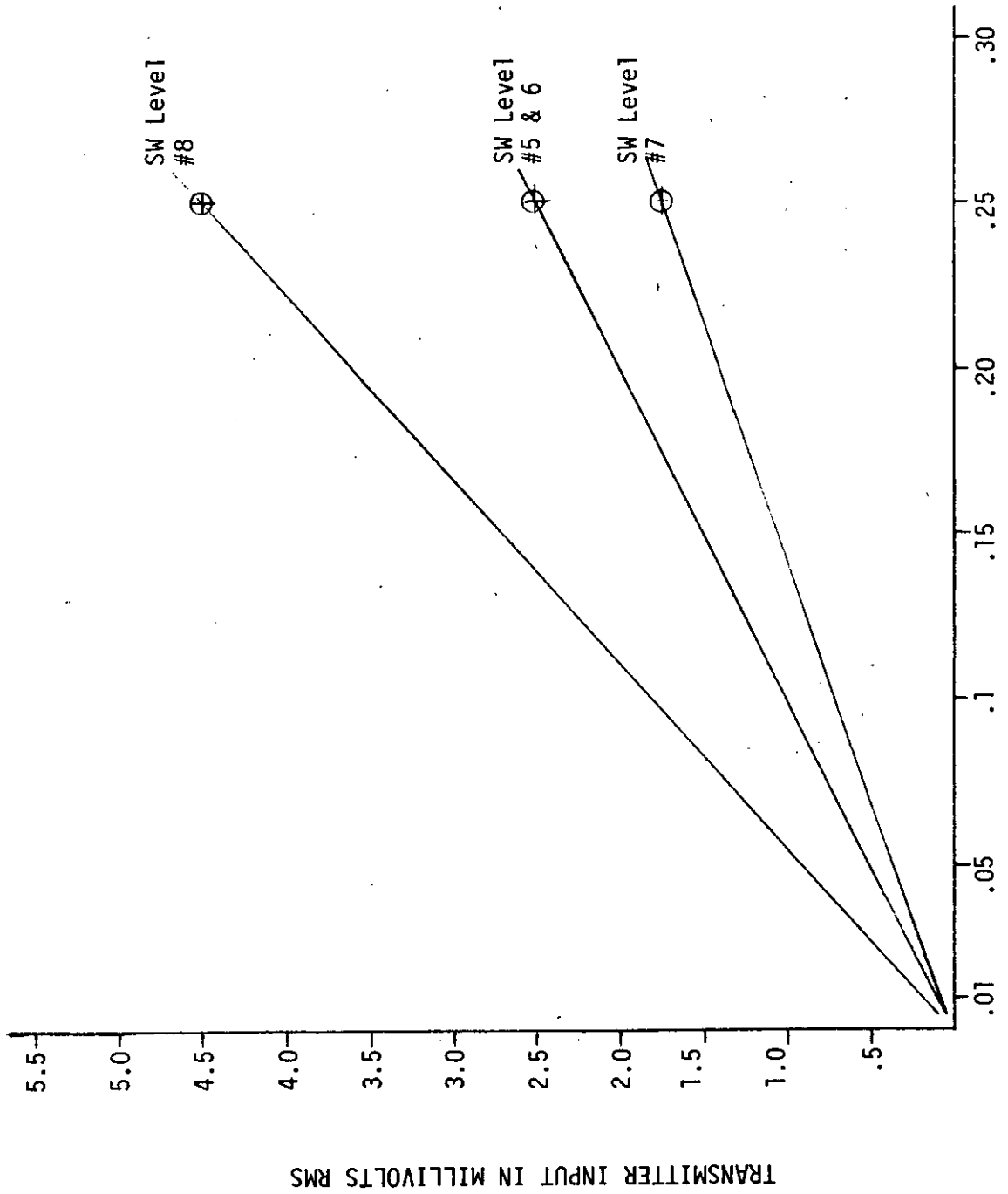
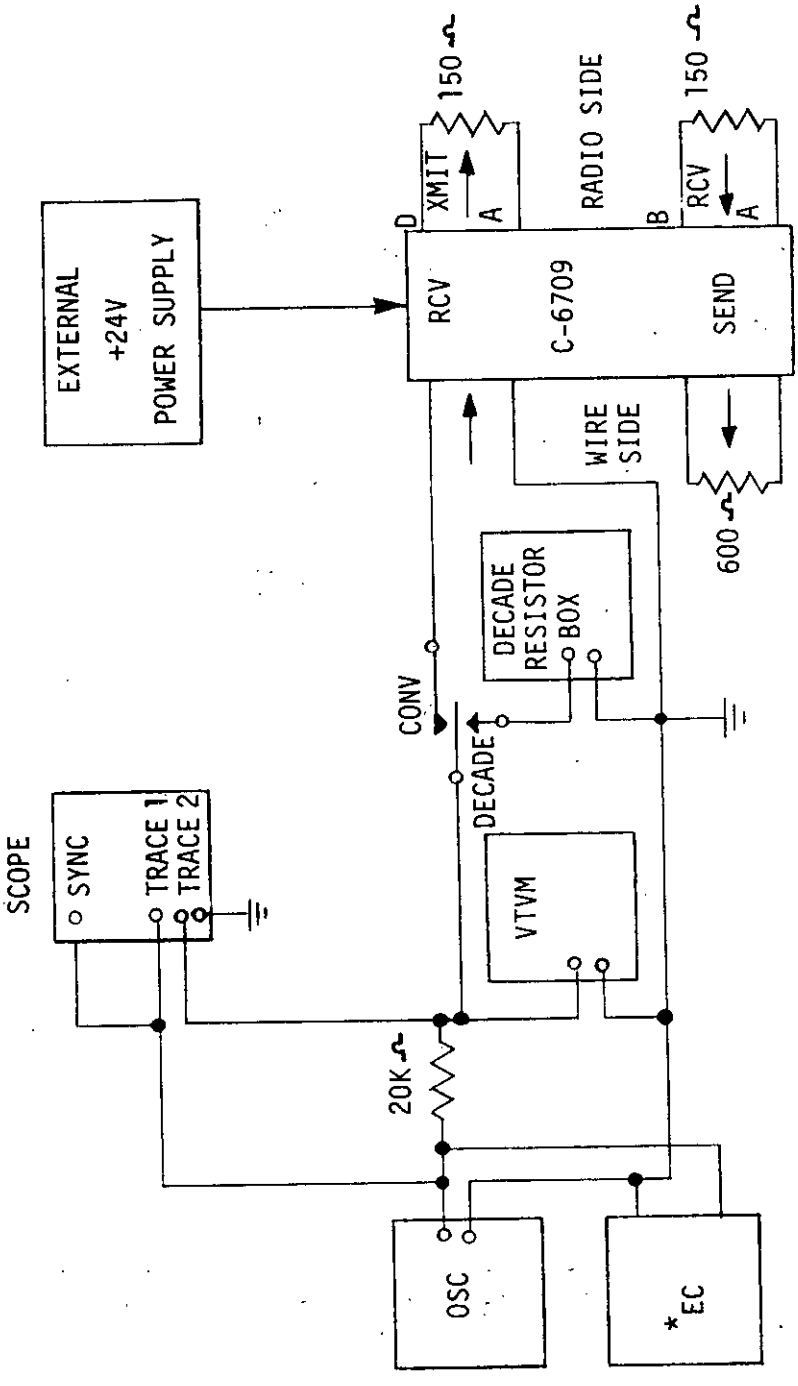


FIGURE 3

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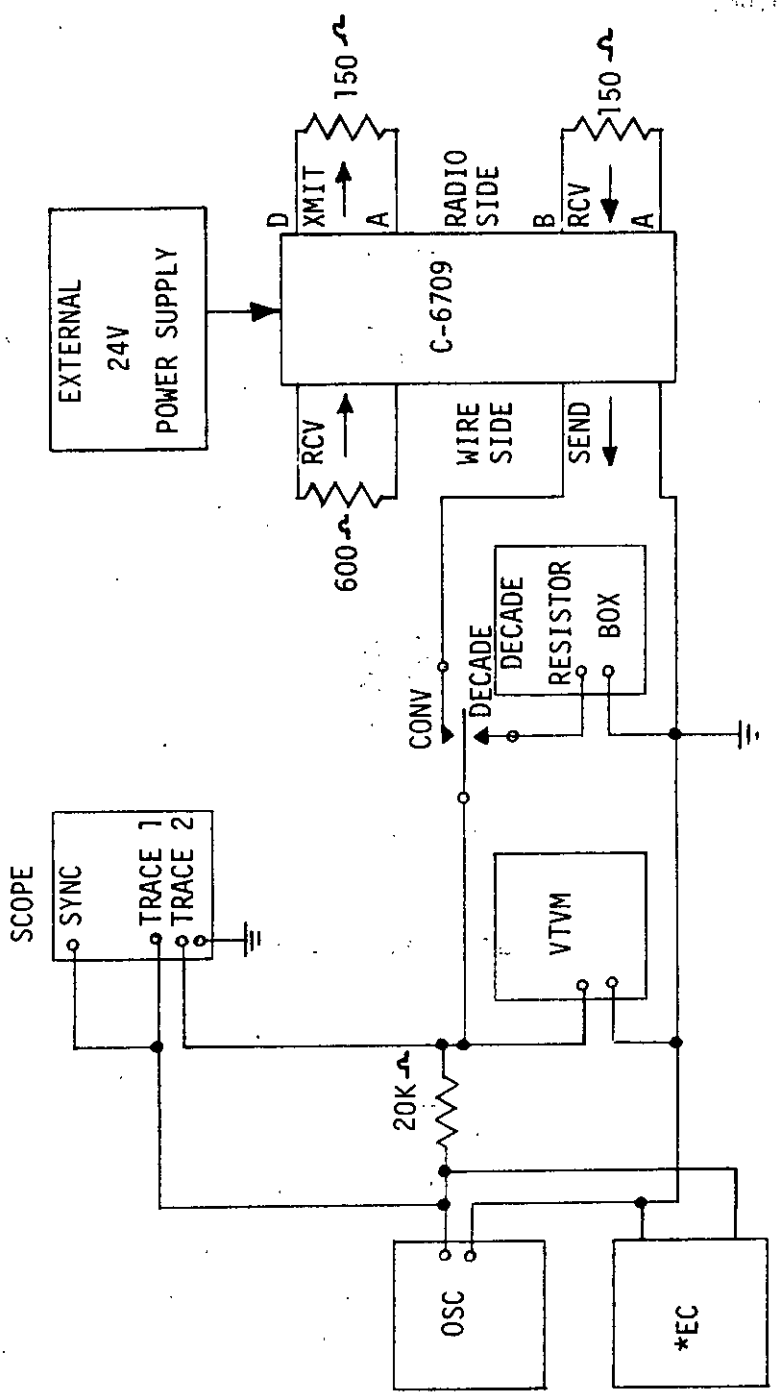


IMPEDANCE TEST SETUP (WIRE REC SIDE)

FIGURE 4

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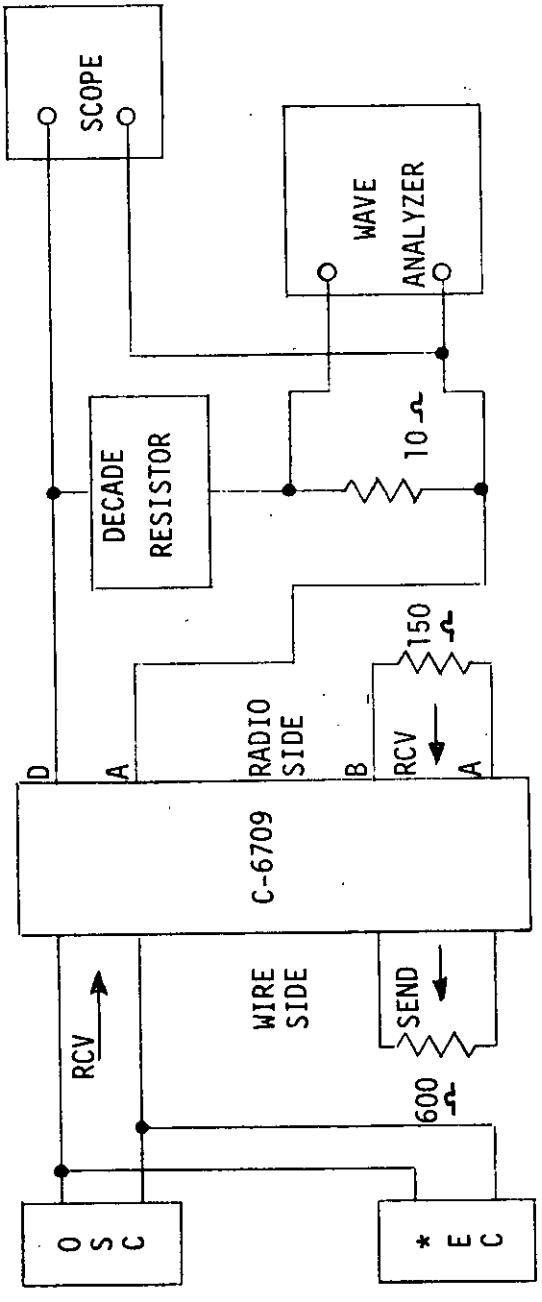


IMPEDANCE TEST SETUP (WIRE SEND SIDE)

FIGURE 4a

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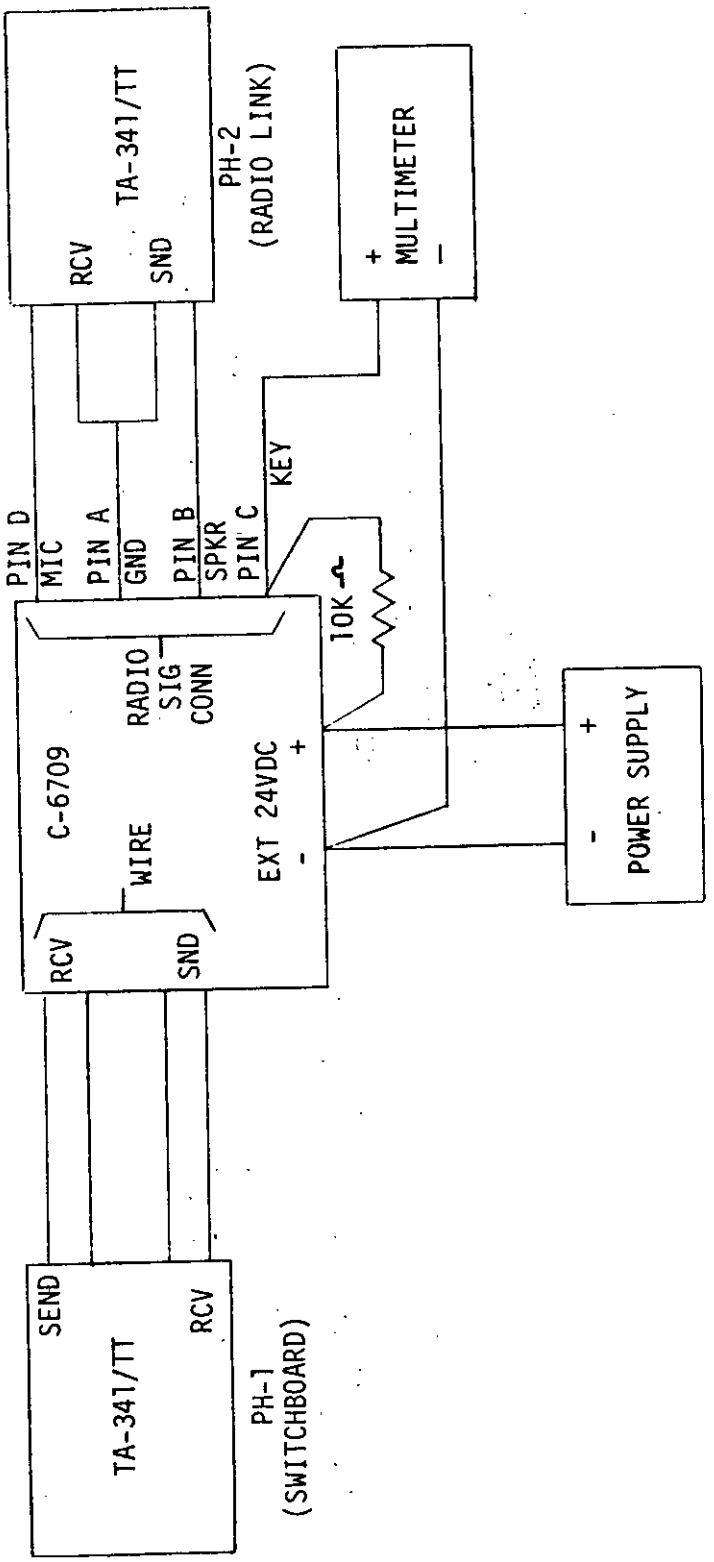


IMPEDANCE (RADIO SEND SIDE)

FIGURE 5

*Electronic Counter

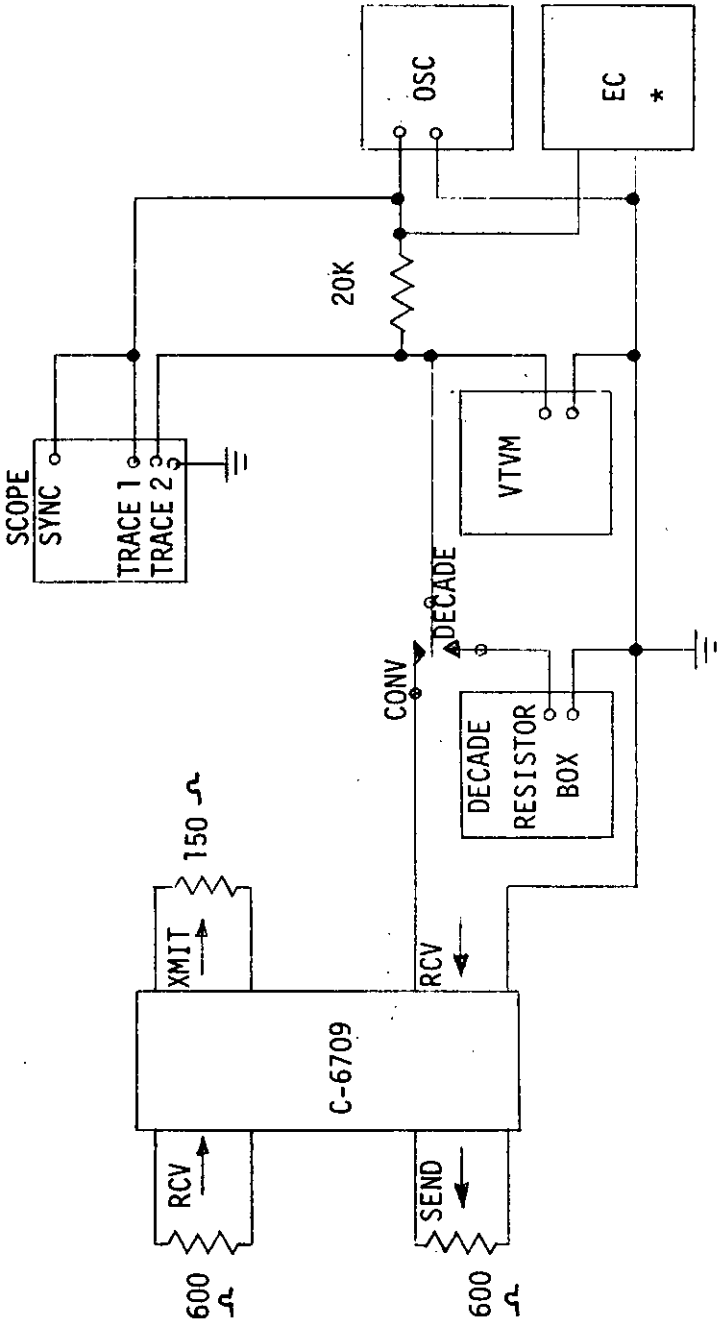
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OPERATIONAL TEST SETUP

FIGURE 5A

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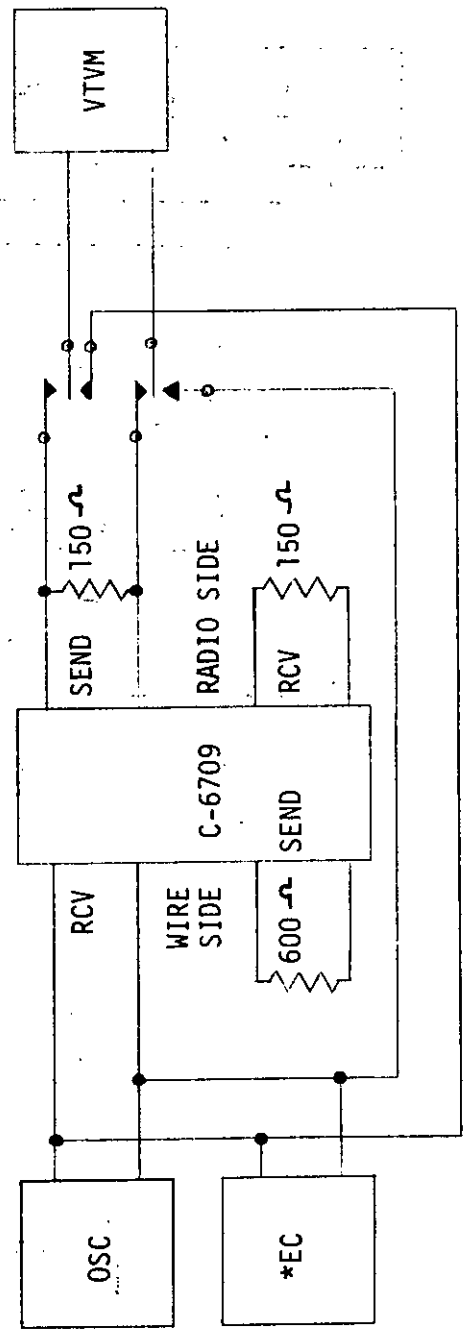


IMPEDANCE (RADIO REC SIDE)

FIGURE 6

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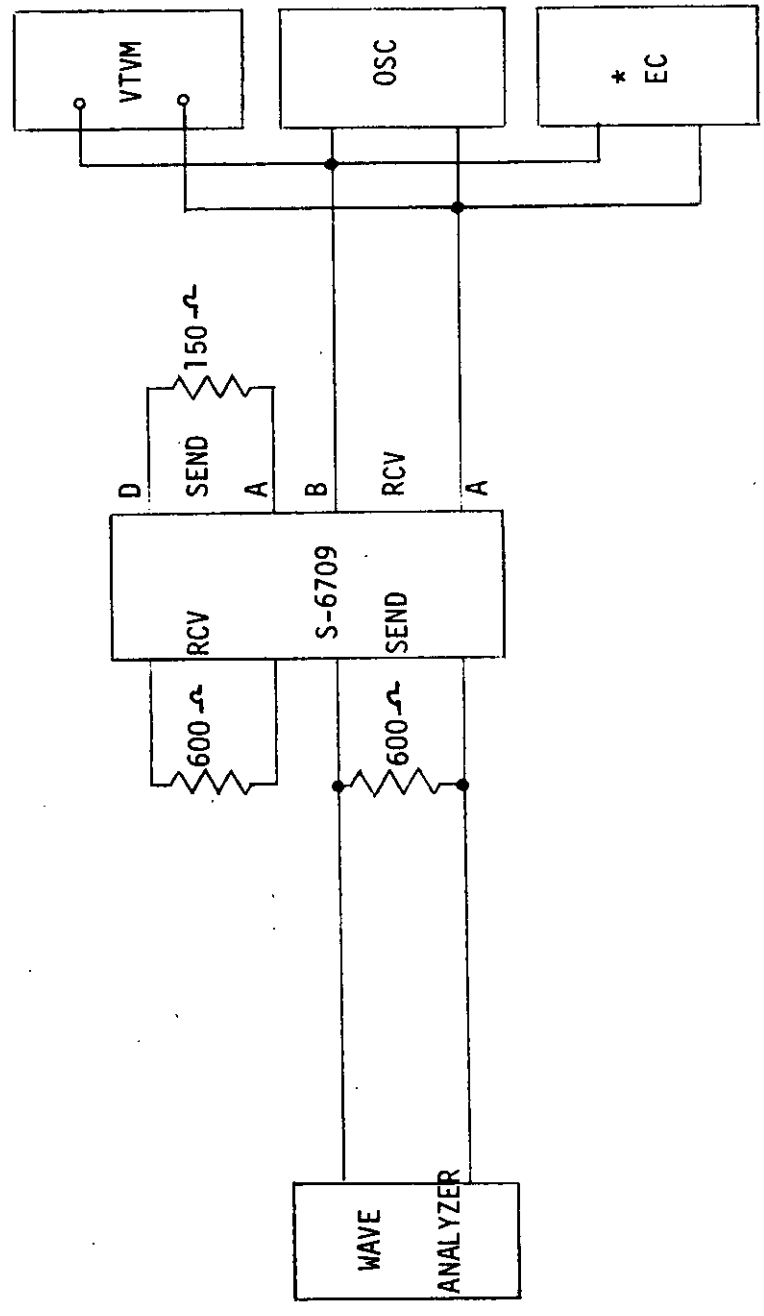


FREQUENCY & AMPLITUDE RESPONSE TEST SETUP

FIGURE 7

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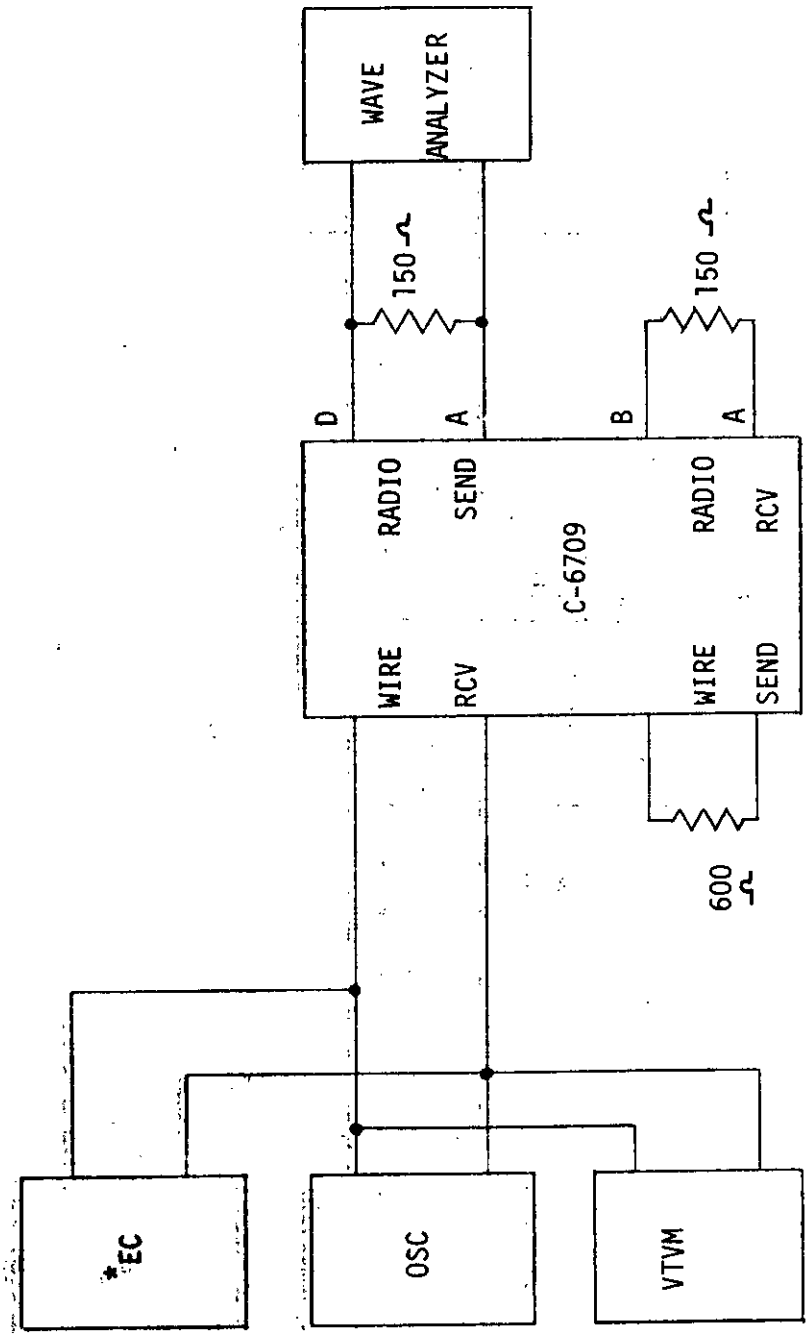


AUDIO SIGNALS TEST SETUP

FIGURE 8

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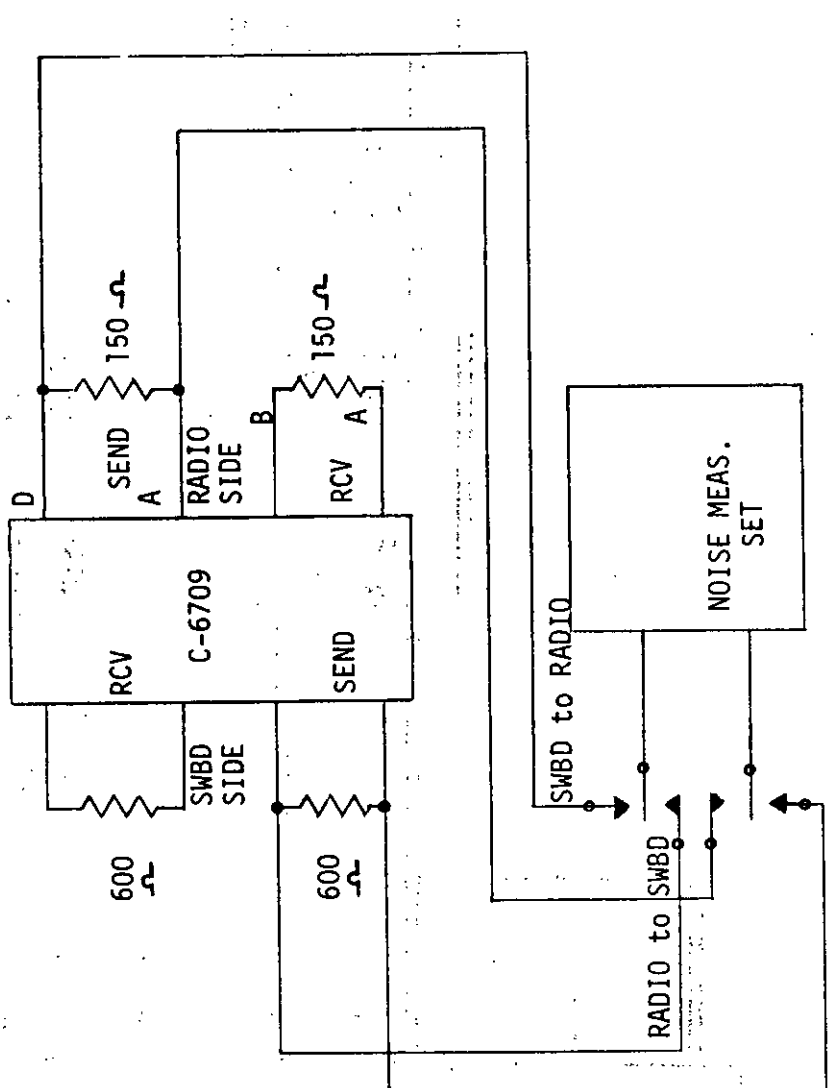


LONGITUDINAL BALANCE TEST SETUP

FIGURE 9

*Electronic Counter

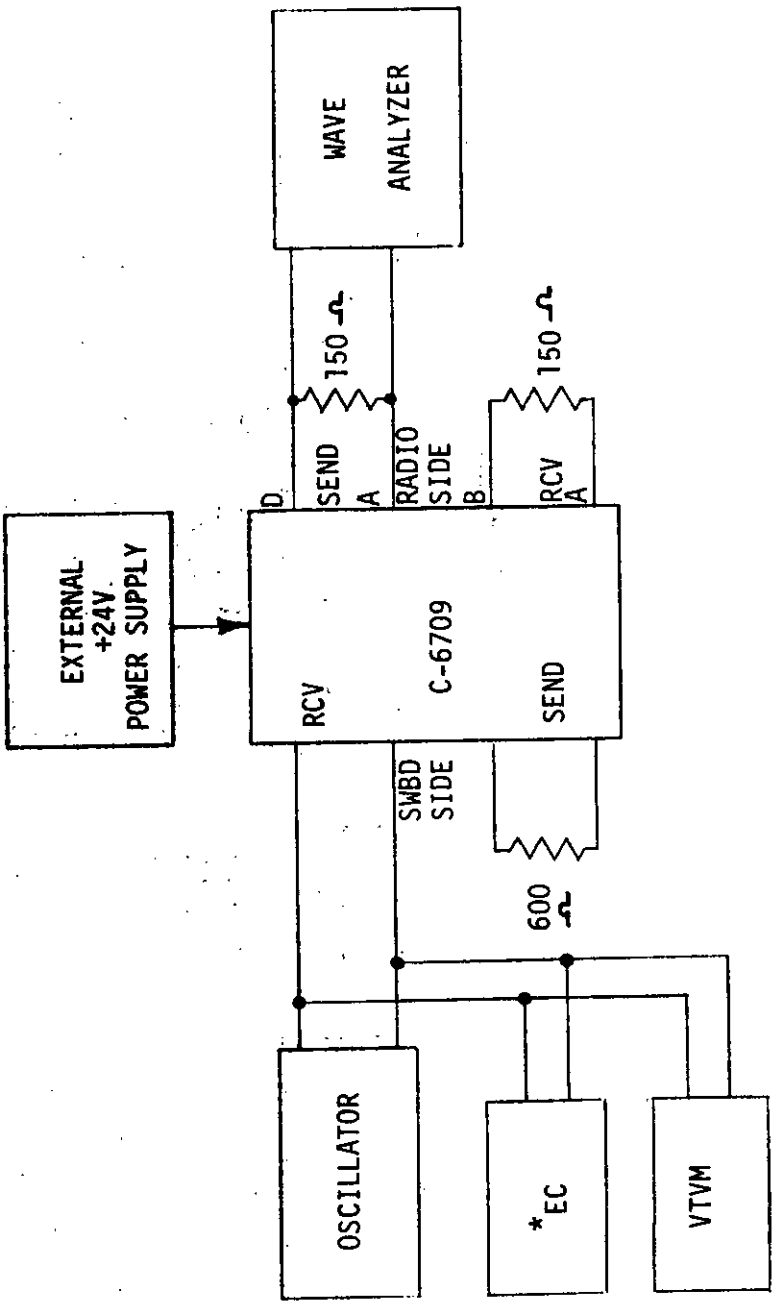
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NOISE LEVEL TEST SETUP

FIGURE 10

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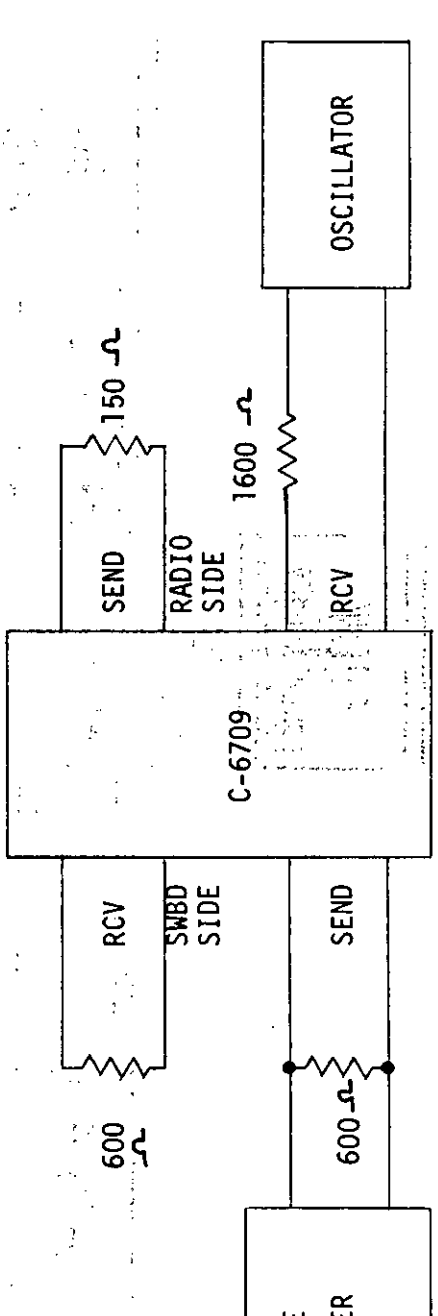


HARMONIC DISTORTION TEST SETUP

FIGURE 11

*Electronic Counter

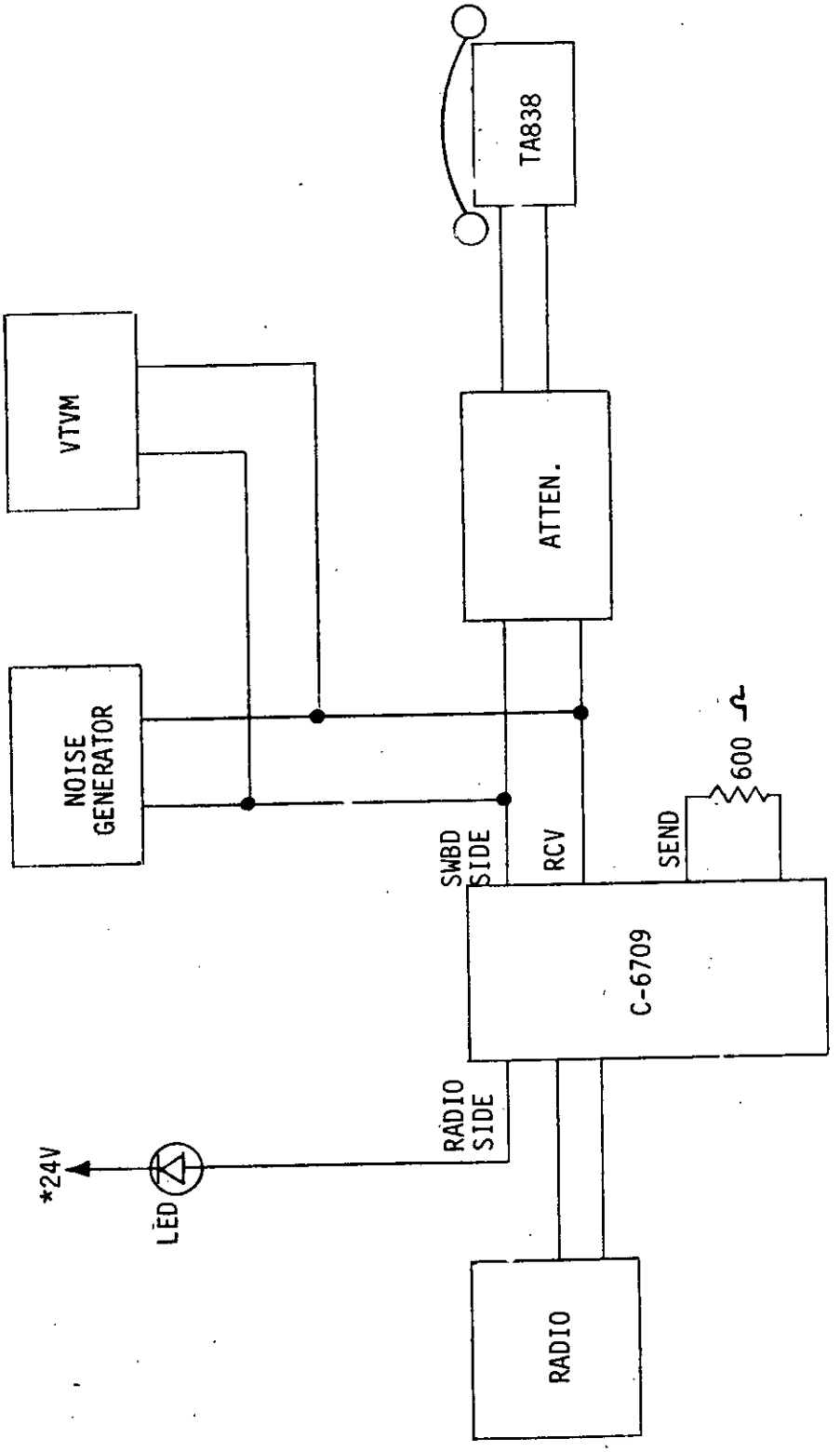
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DISTORTION TEST SETUP

FIGURE 12

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AVOX TEST SETUP

FIGURE 13

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DETECTOR SENSITIVITY

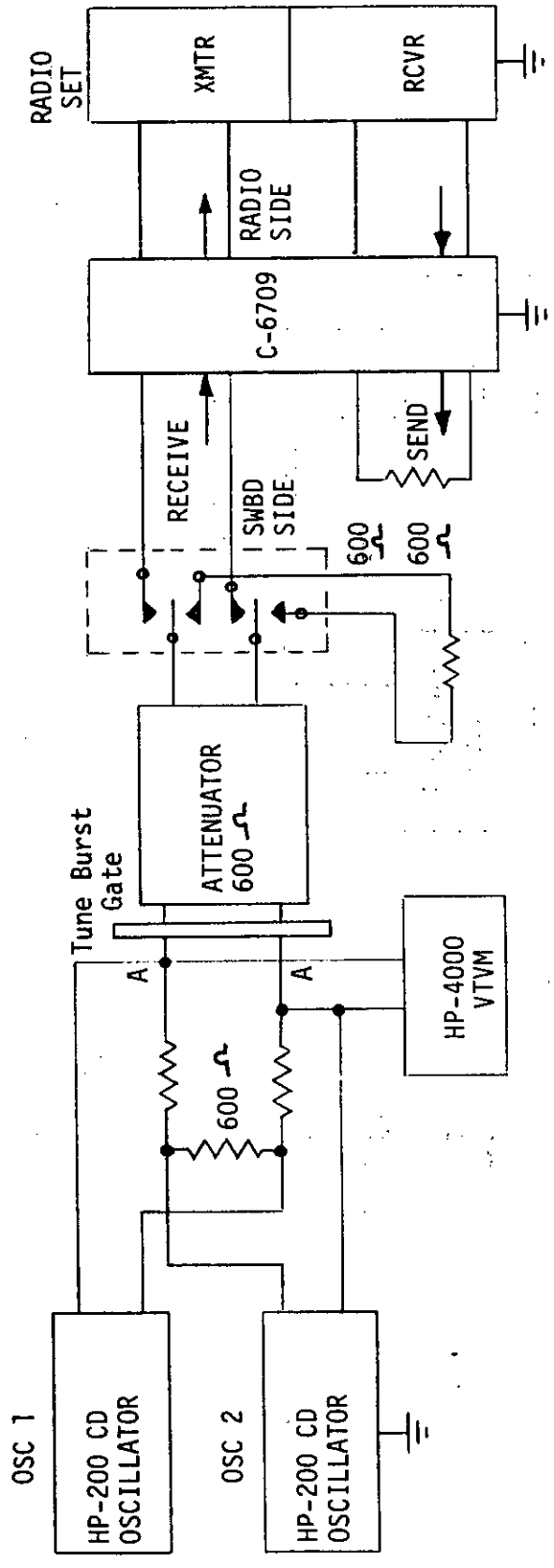


FIGURE 14A

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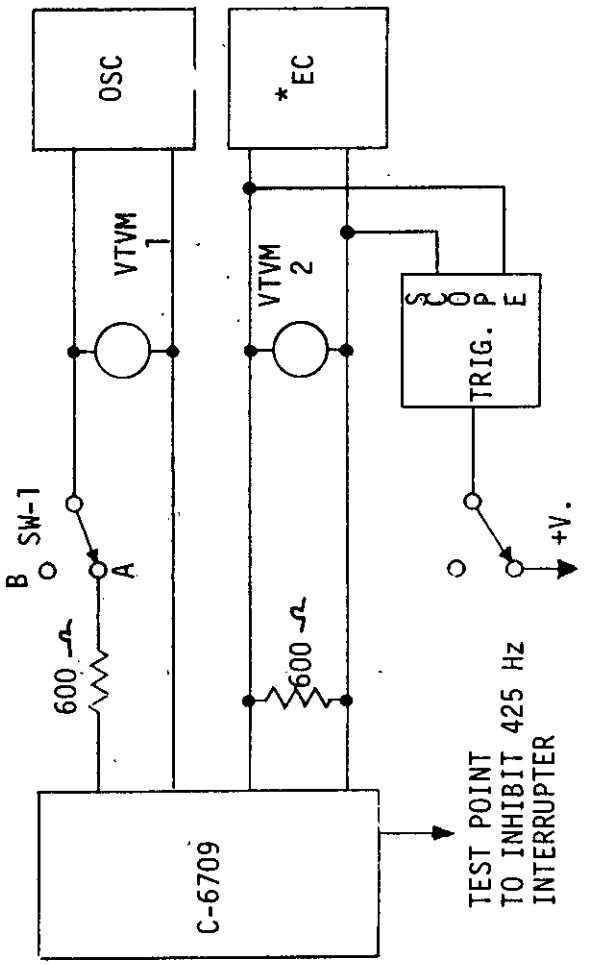


FIGURE 14B

*Electronic Counter

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AUDIBLE ALARM & HEADSET TEST SETUP

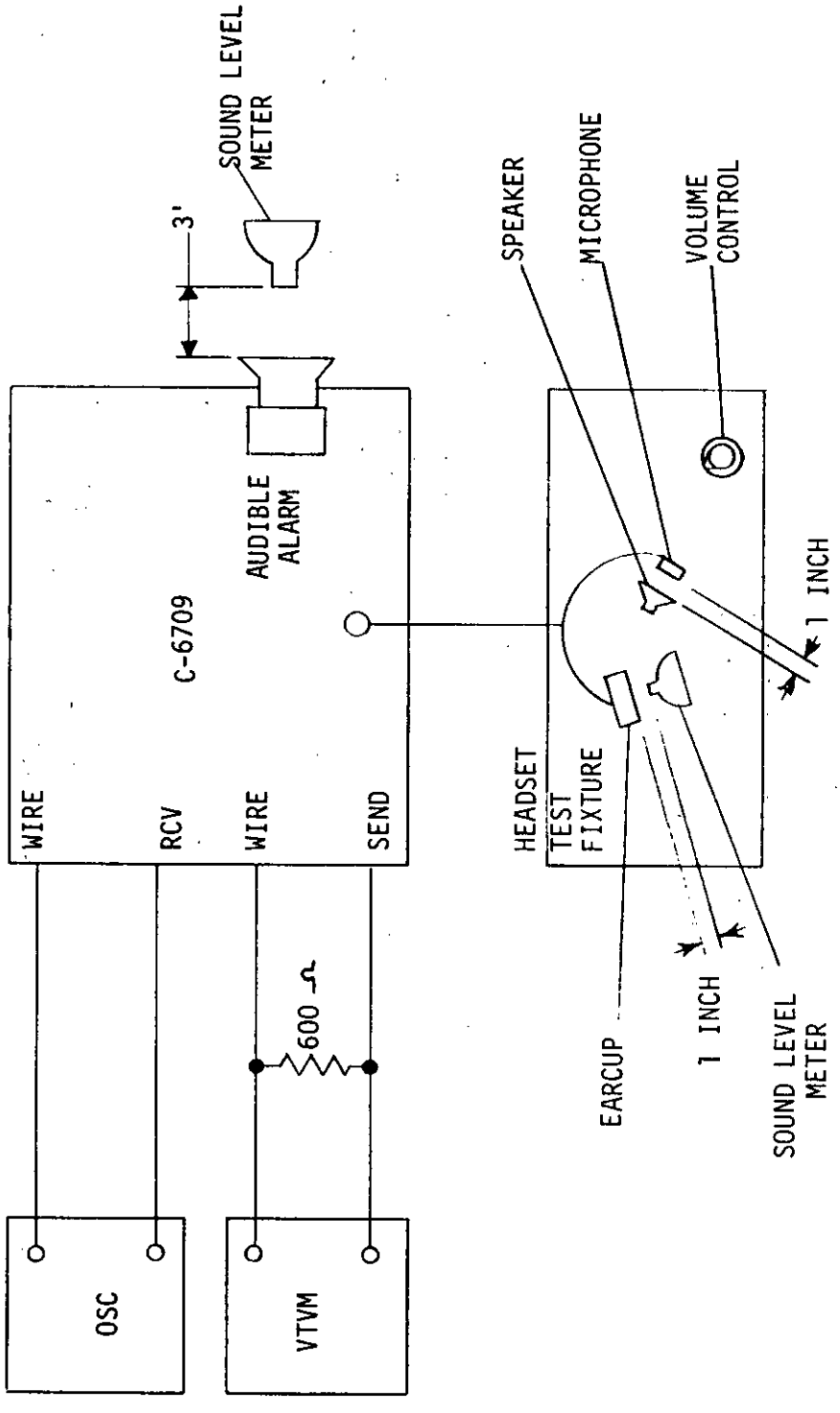


FIGURE 14C

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